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A systematic review of urban vitality studies: Trends and research opportunities

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Abstract: Urban vitality serves as a crucial gauge of sustainable city development. However, little is known about trends and potential research opportunities in urban vitality. Thus, this paper systematically reviews urban vitality studies to synthesize research themes and identify key research directions. Specifically, we provide an overview of the main topics, methods, and data in related studies, synthesizing findings and pinpointing areas for further investigation. Employing the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) method, we identified 191 relevant studies from 941 peer-reviewed articles. Results revealed that Southern Europe and East Asia emerged as the most extensively researched geographic regions, with China featuring prominently. Our review underscores urban vitality measurement (36.7%) and its associated factors (63.3%) as primary research topics. It also highlights a tendency in current studies to overlook assessment biases stemming from urban functions and seasonal variations. While the relationship between urban vitality and the built environment receives ample attention, the impacts of urban vitality on sustainability (environmental, social issues, pandemic) remain less elucidated. Furthermore, we observed that previous studies of influencing factors on urban vitality often rely on linear models, neglecting nonlinearity, interaction, and causality. We also note a shift from traditional data sources to big data, allowing scholars to assess urban vitality comprehensively. We discussed potential research directions for future studies. This paper

advances understanding of urban vitality research by delineating trends, gaps and identifying avenues for future studies.

Keywords: urban vitality, street vitality, urban vibrancy, PRISMA, systematic review

1. Introduction

Urbanization has been a global trend in recent decades, driven by population growth, economic development, and globalization (Lu et al., 2018). Two-thirds of the world's population is expected to live in urban regions by 2050 (United Nations, 2018). While urbanization can promote city development, it also causes negative consequences, such as ghost cities (Cobbinah et al., 2015; Shi et al., 2020); urban shrinkage (Du et al., 2020; Yang et al., 2021b); environmental degradation (Yao et al., 2022); social problems (Yao and Jiang, 2021); and urban quality deterioration (Li et al., 2023). For instance, auto-oriented development has led to rapid urban expansion in the U.S., but density in central areas has decreased in some cities (Barrington-Leigh and Millard-Ball, 2015). In China, recent emerging phenomena of urban decline have prompted researchers to find solutions to revitalize urban areas. As an indicator of a city's internal development force, urban vitality measures its attractiveness, vibrancy of social activities, and development quality (Lan et al., 2020; Yue et al., 2021). Urban vitality creates a driving force for sustainable economic development to maintain vibrancy and is considered a competitive energy source for the city. Therefore, expanding the understanding of urban vitality is integral to sustainable urban planning.

Urban vitality is a concept that has been used widely since the 1960s. Initially introduced by Jacobs (1961), urban vitality refers to the dynamic and diverse urban life encompassing human activities and living spaces. Over time, the term's meaning has involved maintaining internal growth and development, which developed from a good city form (Lynch, 1984). Maas (1984)

linked urban vitality to spatial quality that derives from diverse distinctive business and entertainment opportunities and a substantial number of socially diverse pedestrians. Meanwhile, Montgomery (1998) used the term urban vitality to refer to the extent to which a place feels alive or lively. According to Alexander (2002), a vibrant city is a properly planned and legible form that promotes interaction and connects human activities. These varied definitions share a common idea of people's interactions in the streets, public spaces, and other built environments that generate social activities. Nevertheless, despite the concept's frequent use, there is no consensus on quantifying urban vitality.

Urban vitality is widely researched, and scholars measure it on various scales. Scholars such as (Yue et al., 2019a; Dong et al., 2021; Pan et al., 2021; Gómez-Varo et al., 2022) have mainly focused on the built environment aspects of urban vitality, whereas others such as Yue et al. (2021); Wang et al. (2022e) employed human activity and interaction to measure urban vitality. To better understand urban vitality, scholars used a sustainable development approach that considers economic, social, and environmental (Shi et al., 2021), cultural (Li et al., 2022c) and virtual factors (Kim, 2018) in urban vitality measurement. Based on Jacobs's theory, urban vitality is expressed through density, accessibility, diversity (Delclòs-Alió and Miralles-Guasch, 2018), and mobility (Liu et al., 2022a). Moreover, vitality is diverse in terms of space (transit-oriented development -TOD, residential project, street block, and village) and time (e.g., day, night, and time series (Kim, 2020b)). Through various lenses, the vitality analysis has provided comprehensive spatiotemporal trends of urban dynamics. Additionally, the availability of a wide variety of big data has enabled researchers to quantify urban vitality more comprehensively. The data sources include points of interest (POI) (Liu et al., 2020; Yue et al., 2021); mobile phone data

(Liu et al., 2019; Kim, 2020b; Guo et al., 2021); smart cards (Dong et al., 2021; Tu et al., 2022); and other mobility datasets (Zeng et al., 2020; Li et al., 2022c).

Still, the relationships between urban vitality and various aspects of urban life continue to be explored, including the built environment, human activity, and socioeconomic factors. Regarding the built environment, land use diversity, accessibility, block size, and building density were closely linked to the vitality of US cities (Jacobs, 1961). Subsequent studies have confirmed the roles of junction density (Long and Huang, 2019), old buildings, and transport accessibility in promoting urban vitality (Lu et al., 2019; Wu and Niu, 2019; Fan et al., 2021). Regarding human factors, the relationship between urban vitality and human behavior has garnered much attention. Studies have demonstrated a clear association between urban vitality and the amount of time the elderly spend walking (Marquet and Miralles-Guasch, 2015a), physical activities (Lotfata and Helbich, 2022), and human capital (Yang and Pan, 2020). Moreover, a variety of socioeconomic aspects have been explored, including urban innovation (Chen et al., 2022b), real estate (Barreca et al., 2020a), economic landscape (Jia et al., 2021), and green space (Lopes and Camanho, 2013).

Despite the longstanding interest in urban vitality, a notable gap in the literature is the absence of comprehensive and systematic reviews on the subject. One conference review paper by Annunziata and Garau (2021) focuses on the impact of urban morphology on vibrancy and urbanity. Additionally, Jiang et al. (2023) focus on vitality studies in China, whereas Liu et al. (2023c) focus on the contents and methods of vitality assessment at the neighbourhood scale. Urban vitality is a complex theme related to various urban development objectives. Reviewing vitality assessment alone may not cover the full spectrum of urban vitality's interactions with other urban subjects. Besides, urban vitality has been studied worldwide from micro to macro scales, not only at the neighbourhood scale. The present work, therefore, aims to conduct a systematic

review of urban vitality measurement and its determinants through publications worldwide, identify gaps, and suggest future research directions. To achieve the objectives, we address the following three research questions: (1) “What are the characteristics of the studies in urban vitality?”, (2) “What are the dimensions of research on urban vitality, including topics, data sources, and methods?” and (3) “What are promising topics for future research?”. This review provides valuable insights for academia and governments seeking to understand and promote urban vitality.

This paper is structured as follows. Section 2 critically analyzes the evolution of the urban vitality concept and how Jane Jacobs’ foundation works to contemporary interpretation. Section 3 describes the systematic review method and explores research topics through keywords. Section 4 summarizes findings from previous studies. Section 5 critically analyzes the strengths and limitations of related research and outlines future research directions, and Section 6 concludes the study.

2. Evolution of urban vitality concepts

The concept of urban vitality finds its foundational roots in Jane Jacobs' seminal work, *The Death and Life of Great American Cities* (1961), where she envisioned a city that was human-scaled and socially vibrant with daily interactions of street life. Jacobs articulated four key "generators of diversity" essential for vitality: (1) mixed land uses that attract diverse people and activities, (2) short blocks that increase pedestrian permeability, (3) buildings of varying ages to ensure economic diversity and affordability, and (4) concentration of people and buildings to support street-level activity. She further emphasized walkability to foster vibrant streets and cautioned against "border vacuums" that stifle vitality. These insights were grounded in direct observation and a normative vision of inclusive, active urban life.

Jacobs' ideas marked a significant transformation from the top-down, modernist planning paradigms of her time and inspired a wave of empirical and theoretical work. In the 1970s and

1980s, scholars such as Whyte (1980) expanded on Jacobs' observational approach, linking vitality with aliveness and activity density. Whyte used film and behavioral mapping to analyze how people engage with public spaces. Lynch (1984)'s work on cognitive maps and spatial legibility also contributed to understanding how urban form contributes to perceived vitality. Around the same time, Maas (1984) continued to expand Jacobs' idea by emphasizing the importance of heterogeneity of users, viewing it as a synergistic outcome of diverse and unique commercial and cultural opportunities coupled with a dense and socially heterogeneous pedestrian presence. During the rise of New Urbanism in the 1990s, Jacobs' principles were formalized into planning frameworks emphasizing compactness, walkability, and mixed-use zoning (Southworth, 2003). However, this period also saw critiques emerge related to a lack of empirical evidence. Scholars also pointed out that Jacobs' framework underemphasized structural inequalities, racial segregation, and gendered experiences of public space. Despite such criticism, her concept of vitality has been enduring for good urbanism and sustainable city development. In response, urban sociologists began integrating vitality with broader discussions of social justice, exclusion, and the politics of public space (Steil and Delgado, 2019). Smith et al. (1997) introduced a more values-based interpretation, associating urban vitality with six key qualities: livability, character, connection, mobility, personal freedom, and diversity—qualities that extended the concept beyond physical form to include civic and experiential dimensions. Landry (2000) further deepened this perspective by insisting that vitality be understood through the interdependence of the social, economic, and physical environments. According to this view, a vibrant city is not only a place of intense activity but also one where legibility, multifunctional space use, and human interaction are tightly interwoven (Alexander, 2002). Montgomery (1998) and Alexander (2002) added an experiential and spatial logic to the idea, seeing vitality as dependent on the interaction between physical places

and human behavior, measured in terms of the richness of movement, sensory stimuli, and use diversity. From an economic aspect, Dougal et al. (2015) reframed urban vitality as a product of endogenous "spillovers" among residents, workers, and firms, connecting vitality to agglomeration economies. Although these approaches enhanced methodological rigor and enabled comparative studies, they often risked conceptual reductionism.

The emergence of big data and computational methods in the 2010s introduced a wave of data-driven frameworks. While Jacobs (1961) conceptualized urban vitality through the lens of social inclusion, human scale derived from lived experience and street-level observation. However, contemporary studies increasingly reduce her complex, multidimensional vision to quantifiable indicators—density and movement. Urban vitality is often inferred from single digital traces such as mobile phone signals, social media check-ins, or GPS trajectories (Tu et al., 2020; Gao et al., 2023), which prioritize spatial and behavioral correlations while overlooking the social interactions and cultural meanings vital to Jacobs' framework. Additionally, many empirical-analytical frameworks emerged, seeking to quantify Jacobs' vitality idea through spatial and morphological indicators such as land use mix, street connectivity, or building density (Delclòs-Alió and Miralles-Guasch, 2018; Paköz et al., 2022). While useful for cross-comparative analysis, this tendency towards quantification also led to an accumulation of empirical work that sometimes reinforced design orthodoxies without critically interrogating urban life's sociocultural, interaction, and experiential dimensions. Moreover, vitality has been inconsistently defined across studies—ranging from pedestrian flows (Xu and Chen, 2021), night-time light intensity (Xia et al., 2020), others on business density (Chen et al., 2022b; Xiao and Liu, 2023), and still others on subjective perceptions (Wu et al., 2023a)—making synthesis difficult and theoretical advancement uneven. Additionally, these concepts have limitations regarding the spatiotemporal aspect highlighted by

Jacobs. In many studies, for example, urban vitality only refers to the intensity of crowded activities at a certain moment. Most empirical analyses ignored diversity and consistency in conceptualizing vitality over time, which are comprehensive dimensions in the urban vitality concept as proposed by Jacobs. Recently, Huang et al. (2019) proposed a framework grounded in empirical metrics, defining vitality as the intensity of social activity, economic engagement, and pedestrian movement. This approach reflects the increasing influence of big data and quantification in urban studies. In a recent study, Liu et al. (2022a) examined urban vitality from the perspective of urban organisms to assess urban vitality by drawing an analogy between cities and living organisms. They propose that a city's vitality hinges on three interrelated components—growth, diversity, and mobility—collectively forming the "vital triangle."

In summary, the urban vitality concept is diverse and not unified because of the cognitive background of scholars. Nevertheless, the common ground among these concepts is a city's capacity, which is fostered by interactions between people and places and is illustrated through density, diversity, and connectivity.

3. Systematic literature review

3.1. Method

This research employed a systematic literature review, as the research questions aim to identify trends and gaps in urban vitality studies. The preferred reporting items for systematic reviews and meta-analyses (PRISMA) framework were used for article selection. It was initially launched as the quantum oxide research online meeting (QUOROM) in 1999 by an international group for meta-analysis reporting (Moher et al., 1999) and updated in 2009 as PRISMA (Liberati et al., 2009). The PRISMA framework includes 27 checklist criteria for the title, abstract, introduction, methodology, results, and discussion sections (Moher et al., 2009) that help identify the strengths and weaknesses of existing literature. Our study followed the PRISMA framework

for systematic literature review, which includes the following steps: identification, screening, eligibility, and selection (Figure 1).

We used the Web of Science (WoS) database to search for peer-reviewed publications. The study did not set a year limitation for publication to ensure a comprehensive dataset. The study was limited to journal articles and conference papers. Keywords were used, including: ("liveliness" OR "lively") AND ("city" OR "neighbourhood" OR "neighborhood") OR "urban vitality" OR "urban vibrancy" OR "street vitality" OR "street vibrancy" OR ("neighbourhood" OR "neighborhood") AND "vitality" OR ("neighbourhood" OR "neighborhood") AND "vibrancy" OR "city vibrancy" OR "city vitality". A total of 941 papers were extracted (until March 31, 2024).

We carefully reviewed each article's title and abstract. After the screening step, 234 papers remain. During the eligibility process, we conducted a full reading of the 234 papers extracted in the screening stage. We applied the following criteria to select papers for exclusion from our shortlist: (1) the article had not been reviewed previously, (2) the article was not written in English, (3) the article was not relevant to the topic, (4) the article did not have a digital object identifier for access, (5) the article was a review paper, or (6) had weak relevance to the research themes, and (7) short chapters and extended abstracts, and short papers. This round returned 191 publications.

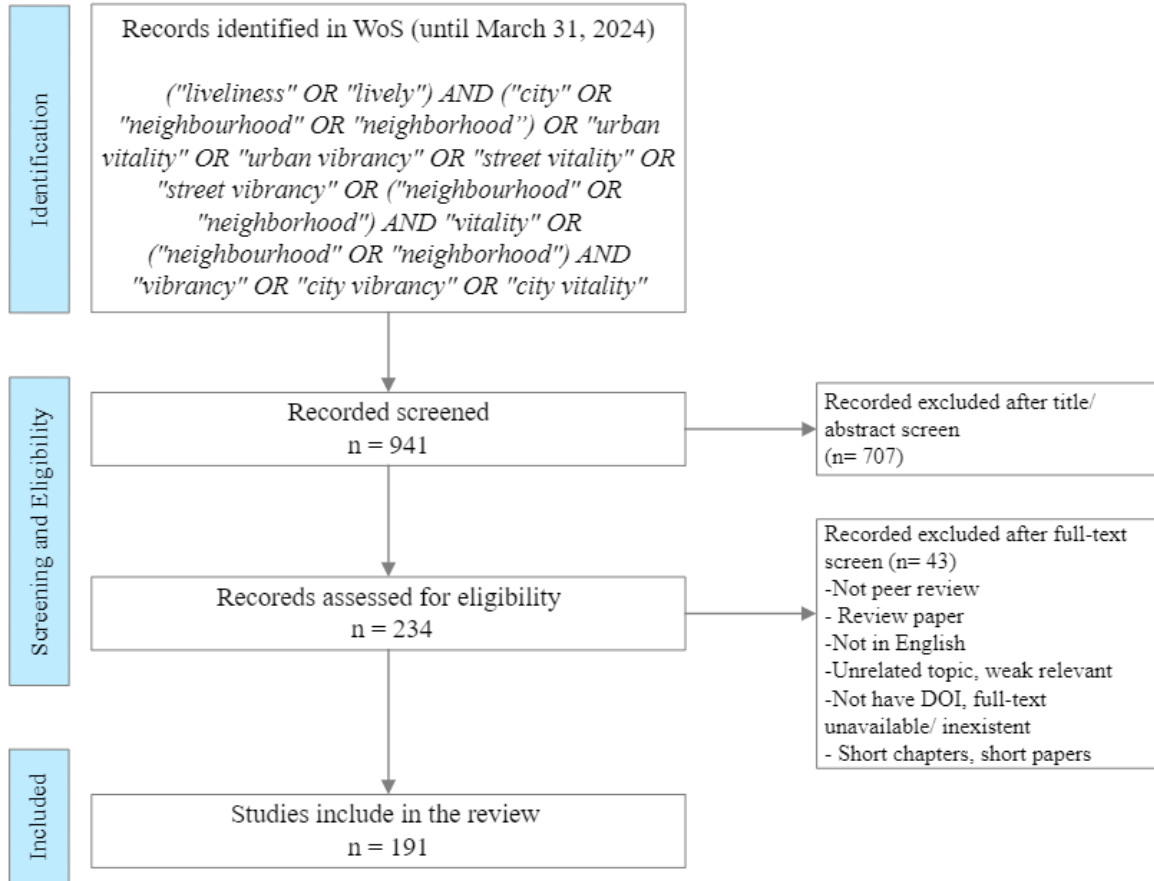


Figure 1. The flowchart of systematic reviews and meta-analyses (PRISMA) method

3.2. Keyword and theme analysis

In academic research papers, keywords serve as markers of the articles' primary content and innovative aspects. Therefore, analyzing keywords can reveal the research trends in research topics. When faced with many keywords, employing tools like VOSviewer can facilitate the examination of keyword co-occurrence, offering comprehensive insights into research clustering. The visualizations generated by VOSviewer depict nodes and linked items, providing clear and intuitive representations of clusters. In this study, VOSViewer was utilized to analyze the distribution of keyword clusters derived from 427 keywords across 191 filtered papers (Figure 2).

Each node in the visualization corresponds to a keyword, with the node's size reflecting its frequency of appearance in publications. Larger nodes indicate higher frequency. The connection between nodes illustrates the co-occurrence frequency of those keywords in publications. The

analysis reveals that “urban vitality-urban vibrancy-street vitality” emerges as the most frequently employed keyword in the body of literature, closely followed by “built environment”. The resultant 39 clusters, delineated by different colours, prominently feature the green and red cluster of “urban vitality-urban vibrancy” as the central node, trailed by the red cluster of “built environment” and the grey cluster of “big data”. This analysis suggests that the primary research focuses revolve around “urban vitality/vibrancy–built environment,” “urban vitality/vibrancy–big data,” and “urban vitality–Jane Jacobs,” aiming to uncover the relationship between urban vitality and the built environment by using big data and to gauge urban vitality through the lens Jacobs’ theory.

Figure 2. Keyword co-occurrence in urban vitality studie

4. Findings

4.1. Spatial and temporal distribution of urban vitality studies

The number of publications on urban vitality has significantly increased over the past decade (Figure 3). Among the 191 selected papers, one was published in 2013, with a subsequent period of low annual publications, ranging from two to three per year, from 2015 to 2017. However, a notable surge in studies occurred from 2018 to 2023, indicating a growing interest in this field and suggesting a promising potential for further research.

In recent years, urban vitality has emerged as a research hotspot in various disciplines, such as urban planning and design, urban geography, urban economics, and urban sociology. It has also been integrated into the local planning strategies of many cities (Patrizia Sulis et al., 2018; Li et al., 2021a). The rapid increase in articles measuring urban vitality in 2018 can be attributed to several factors, including abundant urban big data and the rise of sustainable development movements.

First, the proliferation of extensive and diverse urban big data has become central to many urban studies (Martí et al., 2019). Notably, high-resolution data on urban settings is now widely accessible on a large scale (Botta and Gutiérrez-Roig, 2021). The advent of big data, characterized by its high volume, real-time updates, accurate location information, and open-source accessibility, has provided new opportunities for spatially explicit measurements of urban vitality (Yue et al., 2019b). Alongside big data initiatives, various entities such as local governments, private enterprises, and voluntary programs are diversifying the data sources available to urban scholars engaged in vitality studies (Long and Liu, 2016). Leveraging urban big data, urban vitality studies now require significantly less time and labor than traditional questionnaire data, resulting in a qualitative leap in sample size. For instance, big data initiatives collecting information from mobile phone activities, vehicle trajectories, and public transit usage empower scholars to better quantify

the relationship between urban vitality and the built environment (Batty, 2012; Niu et al., 2022a), capturing more nuances at finer scales (Yue et al., 2017b; Wu et al., 2018c; Botta et al., 2020; Tu et al., 2020). Building upon these technological advancements, urban vitality was examined in major cities, including Seoul, Milan, and Chicago, in 2017; however, there has been limited research on intra-urban vital characteristics within Chinese megacities (Yue et al., 2019b). Notably, there has been a considerable rise in publications related to Chinese cities since 2018.

Moreover, the introduction of sustainable development goals in 2015, particularly with one indicator emphasizing the need to "make cities and human settlements inclusive, safe, resilient, and sustainable" (Nilsson et al., 2016), may have inspired some scholars to explore vitality related topics. This is because urban vitality, demonstrated by its positive role in promoting social interaction (Jacobs, 1961), fostering innovation (Chen et al., 2022b), enhancing urban competitiveness (Brenner and Schmid, 2014), and contributing to city resilience, aligns with the goals of sustainable development (Meng and Xing, 2019). Numerous scholars have argued that urban vitality is intricately interwoven with the principles of sustainable urban development (Shi et al., 2019; Li et al., 2020).

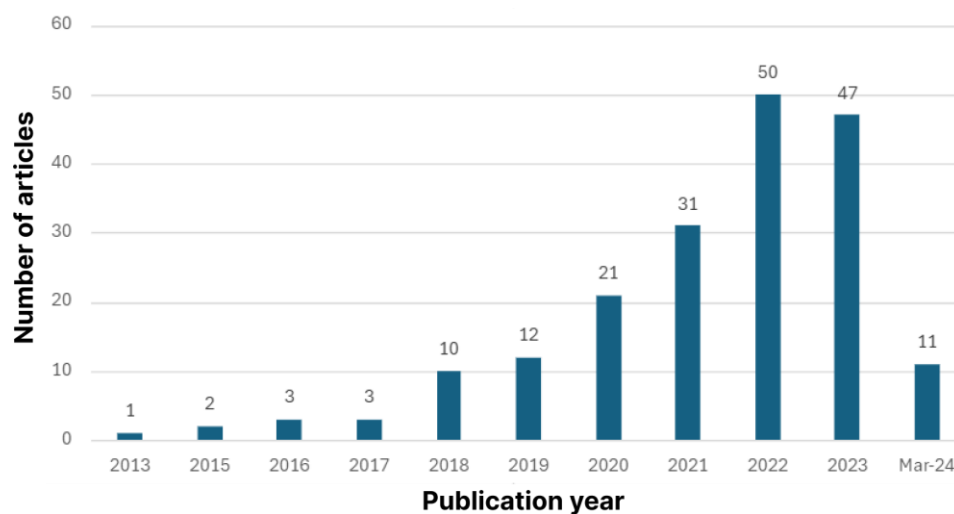


Figure 3. Number of publications of urban vitality studies

Figure 4 shows the spatial distribution of studies in urban vitality, which varies widely across regions. The articles mainly focused on East Asia and Southern Europe, which account for approximately 87.95% and 7.85% of the study areas, respectively. The urban vitality measurement studies mainly focused on populous cities in China and South Korea, such as Shanghai (14 articles), Beijing (8 articles), Nanjing (7 articles), Wuhan (5 articles), and Seoul (2 articles). Similarly, Barcelona (3 articles) and Torino (2 articles) were used as case studies in Europe (Table 1). The relationships between urban vitality and other factors were studied in 36 cities (Figure 4), including 21 Chinese cities. Shenzhen was studied most in 21 articles, followed by Beijing (12 articles), Wuhan (10 articles) and Shanghai (10 articles).

The concentration of studies on measuring urban vitality in major cities in East Asia and Southern Europe can be explained by the urban development context and the availability of mobility data. Firstly, major cities in both regions are experiencing urban shrinkage, evidenced by a decline in urban population density during development (Jiang et al., 2022a). In Southern Europe, cities are witnessing a notable population movement from the urban centre towards the fringe and functional urban areas, with 11.5% of cities experiencing a population decline from 2011 to 2020 (Tsapala et al., 2023). Regional disparities are further highlighted by unemployment rates, with a 5% rate in northern Europe compared to a higher 15% rate in the urban south (Tsapala et al., 2023). In major Chinese cities, the rapid development of new subcenters or towns leads to a decline in urban density in the periphery, spreading urban vitality outward and creating dispersed vitality (Yue et al., 2021), negatively affecting economic resilience (Sun et al., 2023). Moreover, deteriorating infrastructures and subpar living conditions have led to a decline in appeal in numerous neighbourhoods (Tu et al., 2020). These major cities exhibit a unique urban configuration characterized by mixed land uses and high walkability, particularly evident in

Southern European cities with distinct vital nature characteristics (Salvati et al., 2015). They are also distinguished by higher levels of pedestrian activities (Marquet and Miralles-Guasch, 2015b), which is essential for future city development (Zhang et al., 2020a). Rapid urbanization has been a common trend in both regions (Delclòs-Alió et al., 2019; Dai et al., 2022), leading to morphological fragmentation and a greater reliance on high-capacity transportation infrastructures (García-Palomares, 2010). Urban vitality has become a pivotal representation of high-quality city development in these contexts, gaining prominence in urban planning and regional research of big cities. Second, major cities in these regions (Seoul, Tokyo, Shanghai, Beijing, Shenzhen in East Asia and Barcelona, Rome in Southern Europe) also serve as economic, cultural and political centers that draw government and scholarly attention to the dynamics of rapid urban transformation. Additionally, major cities have assumed a prominent position in policy discussions, given their escalating significance in the country's transition and innovation development (Yeh and Chen, 2019). Findings from these cities could serve as a valuable reference for the other city regions experiencing transitioning towards high-quality development (Chen et al., 2022b). Third, human mobility, mobile phones, and activity-related data can be easily obtained in preeminent Chinese cities (Patrizia Sulis et al., 2018; Tu et al., 2020) and Southern Europe major cities (Delclòs-Alió et al., 2019). These data offered novel ways for spatially explicit measurement of urban vitality (Yue et al., 2019b), serving as an intriguing test of the validity of Jacobs' ideas on urban vitality (Delclòs-Alió et al., 2019; Botta and Gutiérrez-Roig, 2021).

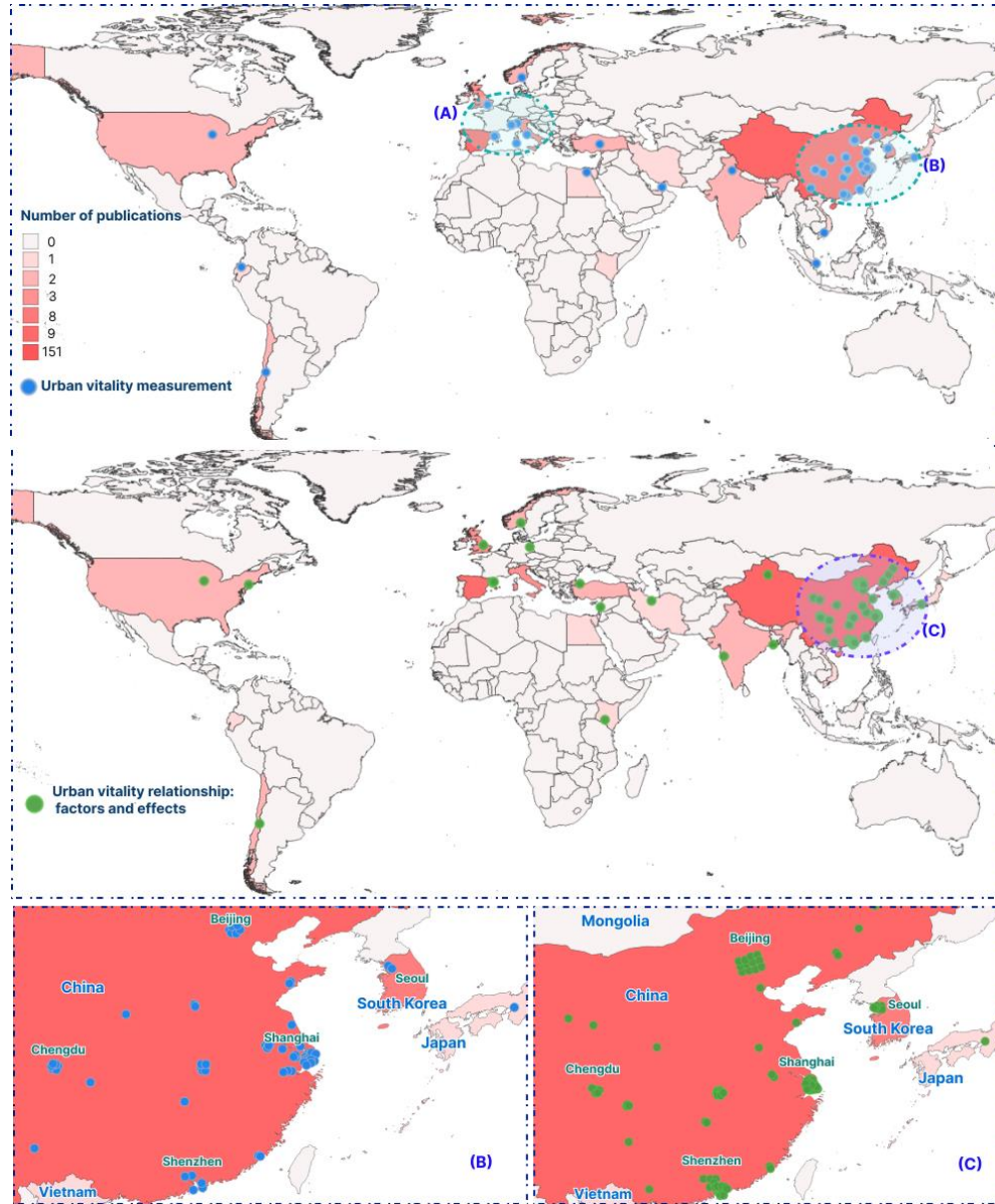


Figure 4. Spatial distribution of urban vitality studies

Table 1. Urban vitality measurement studies in two main geographical clusters

No	City	References
East Asia		
1	Beijing	(Jia et al., 2021; Wang et al., 2021; Zikirya et al., 2021; Huang et al., 2022; Wang et al., 2022f; Wu et al., 2022b; Zhang et al., 2022c; Liu et al., 2023b)
2	Shanghai	(Huang et al., 2019; Yue et al., 2019a; Dong et al., 2021; Liu et al., 2021; Shi et al., 2021; Yue et al., 2021; Zikirya et al., 2021; Li et al., 2022b; Liu et al., 2022a; Wang et al., 2022b; Wu et al., 2022b; Zhang et al., 2022c; Liu et al., 2023b; Zhang et al., 2023b)

<i>No</i>	<i>City</i>	<i>References</i>
3	Wuhan	(Zeng et al., 2018; Fang et al., 2021a; Ou et al., 2021; Wu et al., 2022b; Zhang et al., 2022c)
4	Nanjing	(Xu et al., 2018; Liu et al., 2019; Liu et al., 2020; Liu et al., 2022a; Wu et al., 2022b; Yang et al., 2022b; Hu et al., 2024)
5	Guangzhou	(Zikirya et al., 2021; Niu et al., 2022a)
6	Seoul	(Kim, 2018, 2020b)
<i>Europe</i>		
7	Barcelona	(Delclòs-Alió and Miralles-Guasch, 2018; Delclòs-Alió et al., 2019; Gómez-Varo et al., 2022)
8	Italian cities	(Botta and Gutiérrez-Roig, 2021)
9	London	(Sulis et al., 2018; Tu et al., 2022)

4.2. Research theme analysis

Our examination of keywords and article reviews identified two primary topical categories.

The first category measured urban vitality using urban big data. The second category delved into the mutual relationship between urban vitality and other facets of urban life. Sixty-eight documents (36.7%) were dedicated to quantifying urban vitality. In contrast, 123 papers (63.3%) investigated the connections between urban vitality and other urban attributes.

4.2.1. Urban vitality measurement theme

Previous studies have endeavoured to measure urban vitality across various scales, from macro to micro levels. Among these, the street scale has garnered the most focus, with 66.18% of papers emphasizing the significance of streets and neighbourhoods in evaluating urban vitality (Figure 5), consistent with observations by Jacobs (1961). Only 7.35% of studies explored the sub-district/city level, using the spatial distribution of urban vitality to uncover the city's polycentric structure. Research at this level primarily compares the spatial distribution of urban vitality among city sub-districts. Additionally, 14.71% of studies examined the vitality of specific urban entities, such as metro stations, traditional villages, residential areas, and preservation areas within urban parks.

Specifically, four articles focused on the vitality of metro stations (Dong et al., 2021; Xu and Chen, 2021), open spaces (Niu et al., 2022b; Hu et al., 2024). Four studies explored the urban

vitality of traditional villages (Liu et al., 2022b) and residential areas (Jin et al., 2017; Wang et al., 2021) and historical preservation (Huang et al., 2022), helping to identify areas with low attractiveness, imbalance, or risk of extinction, as well as contributing to ghost city detection. Two studies examined the vitality of urban parks and urban green, which are often regarded as spaces for engaging with nature in urban settings (Banchiero et al., 2020; Dong et al., 2022), highlighting issues that can be addressed to enhance human interaction with the park's green space. Lastly, less than 5% of publications assessed urban vitality at the district level, receiving relatively less attention.

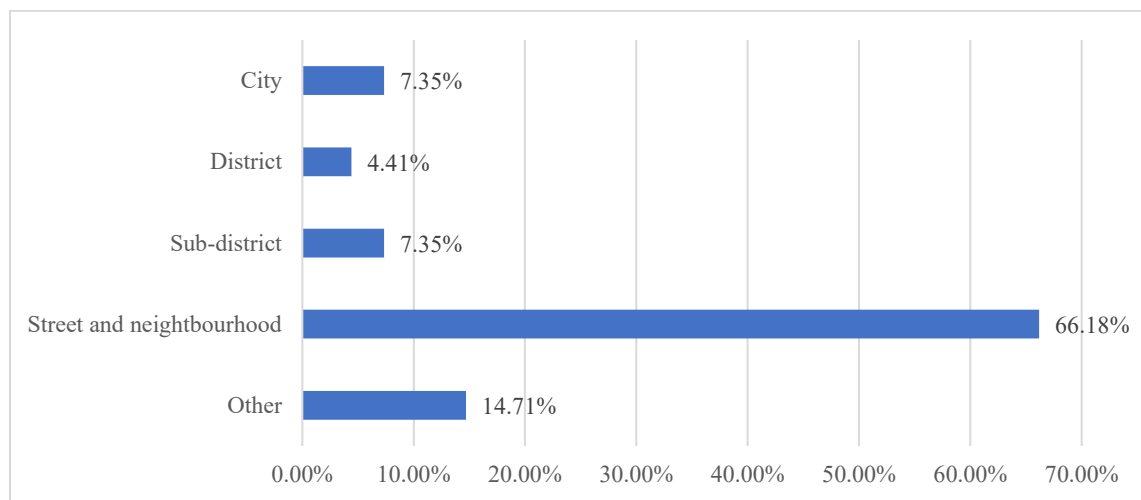


Figure 5. Case study scale of publications

As the environment and activity patterns change, urban vitality can fluctuate over time (Xia et al., 2020). During the day, a city's vitality is often driven by commercial and social activities, such as shopping, dining, and socializing. Parks and public spaces also serve as popular venues for relaxation and recreation during daylight hours. However, as night falls, the city's atmosphere and character may undergo a transformation, with a shift towards entertainment and nightlife. Establishments such as bars, clubs, and restaurants experience heightened activity during nighttime, significantly contributing to the nighttime economy and serving as crucial elements of urban

vitality. Understanding how urban vitality changes over time and how people engage in activities during different periods is crucial for designing and managing vibrant urban spaces. Consequently, many studies have explored the differences in urban vitality by daytime (Li et al., 2022d; Li et al., 2022e; Wang et al., 2024b), nighttime (Wu et al., 2022a; Zhang et al., 2022a; Gong et al., 2023; Yang et al., 2024), and temporal sequences (Yu et al., 2022a; Chen et al., 2023b).

4.2.2. Urban vitality relationship theme: factors and effects

We classify the studies of the relationships between urban vitality and other urban attributes into two categories. The first category includes factors influencing urban vitality, and the second category is mainly about the potential effects of urban vitality (Figure. 6).

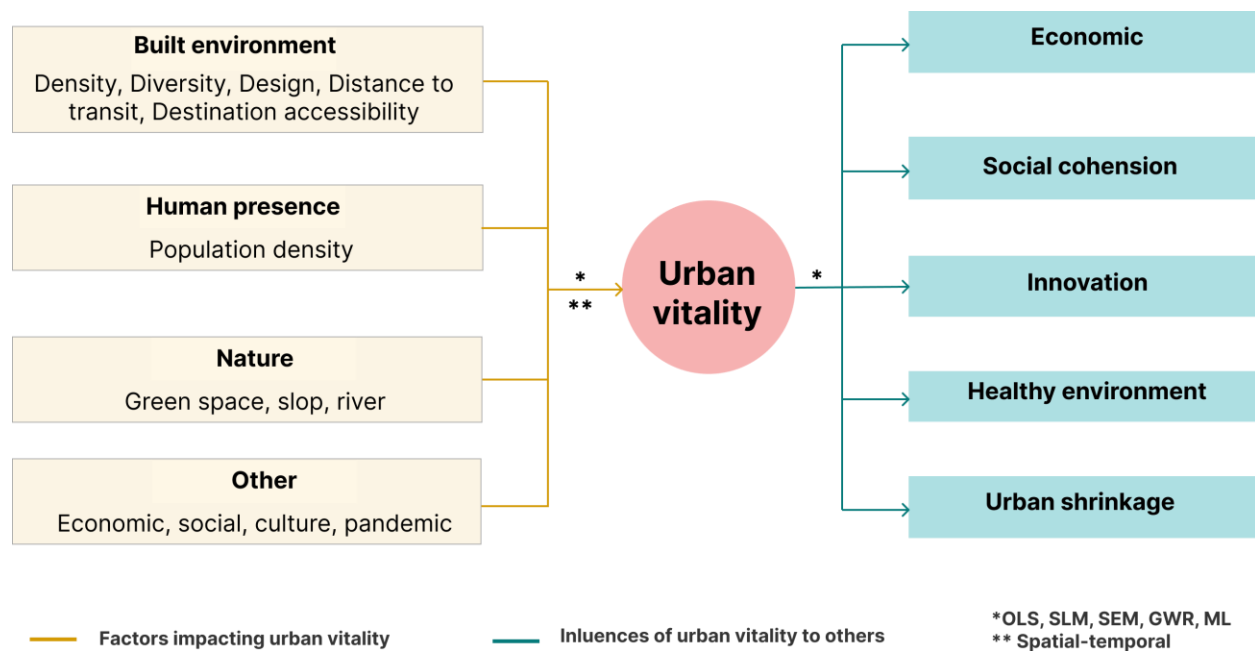


Figure 6. The relationship between urban vitality and associated factors and effects

4.2.2.1. Factors affecting urban vitality

Many factors can affect urban vitality, including three main parts: built environment, natural environment, and human presence. The influence of the built environment on urban vitality

has been a focal point in previous research. In addition to these aspects, scholars have investigated the potential impact of various economic, social, and cultural factors on urban vitality.

a. The built environment

According to empirical quantitative studies, classical urban design and planning theory, the built environment significantly impacts urban vitality. Most studies have attempted to test the association between urban vitality and the built environment factors (He et al., 2018; Wu et al., 2018b; Lu et al., 2019; Meng and Xing, 2019; Quanhua Hou et al., 2019; Wu and Niu, 2019; Yue and Zhu, 2019; Kang, 2020; Kim, 2020a; Mouratidis and Poortinga, 2020; Tu et al., 2020; Yang and Pan, 2020; Zhang et al., 2020a; Chen et al., 2021; Fan et al., 2021; Fang et al., 2021a; Fu et al., 2021; Gan et al., 2021; Guo et al., 2021; Li et al., 2021a; Li et al., 2021b; Liu et al., 2021; Wu et al., 2021; Zhang et al., 2021; Chen et al., 2022a; Dong et al., 2022; Ghosh and Raval, 2022; Gong et al., 2022; Jiang et al., 2022b; Li et al., 2022d; Li et al., 2022e; Liu and Shi, 2022; Lv et al., 2022; Niu et al., 2022a; Tu et al., 2022; Wang et al., 2022a; Wang et al., 2022c; Wang et al., 2022f; Wangbao, 2022; Wu et al., 2022a; Wu et al., 2022d; Xia et al., 2022a; Xia et al., 2022b; Xiao et al., 2022; Yu et al., 2022a; Zhou et al., 2023; Jiang et al., 2024; Li et al., 2024a; Ming et al., 2024). Specifically, we summarised the impact of the built environment on urban vitality based on the 5D built environment framework as follows:

Density: Many scholars have proven that density is an important factor affecting urban vitality (Wu et al., 2018b; Wu and Niu, 2019). Different density factors are found to have associations with urban vitality. It includes block density (Ye et al., 2018), building density (Sung et al., 2013; Lu et al., 2019; Meng and Xing, 2019; Chen et al., 2022a; Wang et al., 2022a; Zou et al., 2023), land use intensity (Fu et al., 2021; Guo et al., 2021; Li and Pan, 2023; Liu et al., 2024a), road density (Yue et al., 2017a; Wu et al., 2018a; Li et al., 2020; Tu et al., 2020; Ding et al., 2023;

Zeng et al., 2023), junction density (Sung et al., 2013; Wu and Niu, 2019), POI density (Meng and Xing, 2019; Li et al., 2020; Jia et al., 2021; Li et al., 2022d; Wu et al., 2022d; Yu et al., 2022a), bus and metro station density (Yang et al., 2021a; Wang et al., 2022g), commercial density (Li et al., 2022e).

Most studies have identified positive associations between density and urban vitality. For instance, building density has been found to positively impact vitality (Meng and Xing, 2019; Wang et al., 2022g). Factors such as compact block form, building variety, building height (Zhang et al., 2020a), and aged building diversity (Wangbao, 2022), positively contribute to residential block vibrancy. Building and interaction density also significantly impact nighttime vitality (Wu et al., 2022a). However, contrasting results were reported by Chen et al. (2022a), who discovered a negative relationship between building density, street intersection density, and urban vitality in Shenzhen. This challenges the conventional notion that building density is a reliable indicator of dynamism in high-density cities. Such findings contradict previous studies on urban vitality in Shenzhen (Ye et al., 2018), suggesting a non-stationary and potentially nonlinear relationship between the built environment and urban vitality. Studies have advocated enhancing shopping, leisure, and bus stop density to promote urban vitality (Li et al., 2022d). The high road and junction density can promote walking activities (Jin et al., 2017; Li et al., 2022d; Lv et al., 2022; Tu et al., 2022; Yu et al., 2022a), and the vitality of urban residential areas is associated with traffic density. Although indicators related to building, road, and intersection density play an important role in various vitality aspects (Chen et al., 2021; Li et al., 2021a; Liu and Shi, 2022), Li et al. (2022d) have pointed out their potential negative impact on urban vitality. Furthermore, scholars have presented evidence suggesting that, in the relationship with urban vitality, diversity may be more influential than density (Li et al., 2021a).

Diversity: The diversity aspect refers to an area that mixes multiple services (Yue et al., 2017a; Wu et al., 2021). It includes facility diversity (Yu et al., 2022b); POI mixed-use (Sung et al., 2013; Tang et al., 2018; Kang, 2020; Li et al., 2020; Mouratidis and Poortinga, 2020; Wang et al., 2022a; Wang et al., 2022g; Huang et al., 2023), land use diversity (Sung et al., 2013; Tang et al., 2018; Kang, 2020; Li et al., 2020; Mouratidis and Poortinga, 2020; Tang and Ta, 2022; Wang et al., 2022a; Wang et al., 2022g; Li and Zhao, 2023; Liu et al., 2024b), mix building age (Li et al., 2020; Wangbao, 2022), function diversity (Zhao et al., 2023).

Overall, there are mixed results on the impact of diversity on urban vitality. For example, Mouratidis and Poortinga (2020) and Chen et al. (2008) considered land use diversity as a strategy for integrating physical functions and social connectivity, enriching infrastructure, and promoting urban vitality (Jacobs, 1961). Sung et al. (2013) and Xia et al. (2020) found a strong positive autocorrelation in the spatial distribution of urban vitality and land use intensity. Additionally, land use mix has a remarkably positive relationship with urban vitality (Mouratidis and Poortinga, 2020; Wangbao, 2022). This effect attracts tourist interest, reduces pedestrian's walking speed, and promotes urban vitality (Zhang et al., 2021). However, Liu and Shi (2022) suggest that controlling for floor area ratio or unintended impacts may negatively correlate land use diversity and urban vitality (Wang et al., 2022f; Xia et al., 2022a). Increasing building function mixed and development intensity in high-intensity areas may not promote street vitality (Wu et al., 2021).

Design: The influences of design aspect were conducted through block size (Li et al., 2020; Gan et al., 2021); TOD (Xiao et al., 2021), road intersections (Fan et al., 2021), floor area ratio (Ye et al., 2018; Lu et al., 2019; Fan et al., 2021; Fu et al., 2021; Jia et al., 2021; Liu and Shi, 2022), building typology (Ye et al., 2018; Jiang et al., 2024), streetlight (Li et al., 2022d), height building (Zhang et al., 2020a; Fu et al., 2021; Yang et al., 2023b), sidewalk (Wu et al., 2021),

pedestrian-friendly street (Li et al., 2021b), number of street lanes (Sung et al., 2013), tree configurations (Fang et al., 2021a; Wu et al., 2022d) such as connectivity, straightness of the road network, the closeness paths, betweenness (Yue and Zhu, 2019), openness and walkability (Lu et al., 2023; Zhao et al., 2023), road width (Istrate, 2023). Street network integration significantly affects urban vitality (Li et al., 2021b; Yu et al., 2022a).

Scholars emphasized the importance of good spatial design to promote urban vitality. In particular, betweenness, followed by proximity and straightness, is most important to urban life when evaluating street centrality in walking mode. However, the correlation between street network characteristics and urban vitality varies on different spatial scales (Fang et al., 2021a). In addition, street width (Hou et al., 2019; Li et al., 2022e) and street aspect ratio (Zhang et al., 2021) are key indicators of differing vitality levels because they influence pedestrian flows. Moreover, block and strip structures positively affect vitality even when building density is considered. Besides, building typologies are associated with urban vitality (Ye et al., 2018). Small block sizes are also thought to facilitate walking and interactions, thus promoting urban vitality (Jacobs, 1961). Sung and Lee (2015) found that urban vitality decreases as the block size increases, which differs from Jacob's theory. However, the impact of block size varies with a non-linear threshold effect on vibrancy (Gan et al., 2021). Additionally, the age of buildings impacts urban vitality, with older buildings positively correlated with street vitality in high-density urban areas (Li et al., 2022d; Wangbao, 2022). However, Sung et al. (2013) found a negative influence of building age on urban vitality. Notably, the floor area ratio is identified as the most significant contributor to urban vitality in the study conducted by Yang et al. (2021a).

Distance to transit: This is an important factor affecting urban vitality. Scholars explored the distance to the nearest highway (Wu and Niu, 2019), transportation facilities (e.g., buses,

subways, and metros) (Sung et al., 2013; Chen et al., 2022a; Tang and Ta, 2022; Wang et al., 2022f; Wang et al., 2022g; Shi et al., 2023; Yang et al., 2023c; Ding and Wang, 2024) are important in promoting urban vitality (Kang, 2020; Xia et al., 2022a). Using Chengdu as a case study, Lu et al. (2019) demonstrated that public transport accessibility positively correlated with neighbourhood vibrancy. Certain factors (e.g., proximity to transportation) positively contribute to urban vitality (Liu et al., 2021; Li et al., 2022d; Yang et al., 2023d). The key features of vibrant blocks are effective transport services and convenient access to subway and bus stations (Zhang et al., 2020a). Subway stations significantly enhance street vitality, while bus services contribute to vibrancy around metro stations (Xiao et al., 2021). Subway stations are considered important entrances to the street, and they coordinate with commercial facilities. Recent research by Fan et al. (2021); and Liu and Shi (2022) also observed the negative impact of distance to the nearest bus and subway station on urban vitality.

Destination accessibility: Destination accessibility is often used in urban vitality relationship studies (Yue et al., 2017a; Lu et al., 2019; Li et al., 2021a; Wang et al., 2022f; Wangbao, 2022; Gao et al., 2023; Istrate, 2023). POI accessibility is widely used as destination accessibility, which includes commercial (Tang et al., 2018), retail, public parks (Wu et al., 2018b), and public places (Wang et al., 2022f). Scholars' findings generally indicated that low accessibility results in declining urban vitality. However, their correlation level may vary locally due to economic statuses and demographic characteristics (Wang et al., 2022g). For example, Xiao et al. (2021) found that POI accessibility has a minor influence on vibrancy around TOD compared to socioeconomic characteristics. Distance to the city center negatively affects urban vitality (Tang et al., 2018; Mouratidis and Poortinga, 2020). Proximity to the park/ plaza negatively influences

urban vitality (Li et al., 2022d). Poor accessibility leads to low street vitality despite the high degree of mixed function (Zhang et al., 2021).

b. Human presence

In previous studies, population density was usually used as a proxy for human presence. There was controversy about the influence of population density on urban vitality. Some studies found a positive impact of population density on urban vitality (Yue et al., 2017a; Jia et al., 2021; Chen et al., 2022a; Paköz and Işık, 2022; Yu et al., 2022a; Wu et al., 2023c), while others observed a negative association (Liu and Shi, 2022). At the city level, Lan et al. (2020) revealed that population density does not significantly influence urban vitality in large and medium-sized cities because the impact of additional population flow on social diversity is not apparent. At the micro-scale, neighbourhood density positively fosters urban vitality (Mouratidis and Poortinga, 2020). Conducting the research from a temporal perspective, Kang (2020) revealed that population density is the most significant factor contributing to urban vitality. Ghosh and Raval (2022) detected both negative and positive relationships between population density and vitality, as population density initially results in urban vitality enhancement and reduction with intense high population density.

c. Natural environment

Studies have explored whether natural factors influence urban vitality, including green space (Lopes and Camanho, 2013; Chen et al., 2022a), topography (Xia et al., 2022a), and rivers (Sung et al., 2013). The results demonstrate that green spaces substantially impact urban vitality by serving as a place for people to engage in physical activity and outdoor recreation (Liu et al., 2021), while border vacuums such as rivers or steepness may disrupt walking activities (Jacobs, 1961; Sung and Lee, 2015), reducing urban vitality. However, Mouratidis and Poortinga (2020);

Fan et al. (2021); Yang et al. (2021a); Chen et al. (2022a); and Li et al. (2022e) found a negative impact of green space on urban vitality. Using NDVI as a representative for greenery, Li et al. (2022d) recognized that increasing NDVI can lead to a decline in urban vitality. On the micro-scale, greenery can increase street vibrancy, while much more greenery can inhibit street vitality on the macro scale as less urban function along the street (Wu et al., 2022d).

d. Other factors

A branch of research has elucidated the intricate relationships between urban vitality and diverse factors, encompassing a healthy environment (Paköz and Işık, 2022); real estate (Barreca et al., 2020b, a); urban shrinkage (Jiang et al., 2022a); reconstruction (Alkazei and Matsubara, 2021); ghost cities (Jin et al., 2017); safety (Humphrey et al., 2019), pollution (Lee and Kang, 2022; Wang et al., 2024a), human perception (Gómez-Varo et al., 2023; Wang et al., 2023; Wu et al., 2023a; Xu and Chen, 2023), covid-19 (Wu et al., 2023b; Xiao and Liu, 2023); urban vulnerability (Gómez-Varo et al., 2024), liveability (Ruszczyk et al., 2023), neighbourhood satisfaction (Madrid-Solorza et al., 2023), and activity, time, and space diversity (Kang et al., 2020a). For instance, the more pronounced the shrinkage phenomenon is, the lower its urban vitality is (Jiang et al., 2022a). He et al. (2018) explore how urban expansion negatively affects urban vitality due to reducing interaction between people and urban environments. Alkazei and Matsubara (2021) have found that poor reconstruction planning leads to a decline in urban vitality as house prices and rent increase. Yu et al. (2022a) pointed out the negative impact of housing price escalation on urban vitality.

Social infrastructure plays a pivotal role in fostering urban vitality. Notably, empirical investigations indicate that the ratio of tertiary industry and the proportion of the youthful and middle-aged population exhibit negligible effects on urban vitality. Conversely, a positive

correlation emerges between per capita income and urban vitality, as evidenced in the study by Lan et al. (2020). Financial investment in social infrastructure (Lan et al., 2020), GDP (Wang et al., 2022f), and income (Tu et al., 2022) have a significant correlation with urban vitality.

4.2.2.2. Effects of urban vitality

We observed a limited body of literature addressing the influence of urban vitality on various urban dimensions. Existing research has explored several relevant themes, such as human behavior (Sung et al., 2013; Marquet and Miralles-Guasch, 2015a; Sung and Lee, 2015; Kang, 2020; Lee et al., 2020; Akinci et al., 2022; Li et al., 2022e; Lotfata and Helbich, 2022; Zordan et al., 2023), social cohesion (Mouratidis and Poortinga, 2020), innovation (Chen et al., 2022b), ghost city (Jin et al., 2017), safety (Humphrey et al., 2019; Lotfata and Helbich, 2022), real estate (Barreca et al., 2020b, a), and pollution (Yu et al., 2022b; Zhou et al., 2023).

Regarding human behaviour, Akinci et al. (2022) indicate that older people tend to rest more in highly vibrant urban areas. Pedestrian volume tends to be higher in densely populated areas with high urban vitality (Lee et al., 2020), where the physical environment is diverse (Sung et al., 2013). Low urban vitality, with a high crime rate and vacant properties, is linked to decreased physical activity (Lotfata and Helbich, 2022). Marquet and Miralles-Guasch (2015a) explored the association between neighborhood vitality and physical activity among older adults.

Regarding social cohesion, Mouratidis and Poortinga (2020) point out the contradiction between urban vitality and social cohesion, as higher vibrant areas tend to have lower social cohesion. However, urban vitality may also positively influence social cohesion by fostering closer relationships with neighbours and increasing neighborly assistance, particularly within similar urban forms.

Additionally, the enduring diminution of urban vitality emerges as a pivotal concern, notably resulting in the manifestation of ghost cities at the micro-scale (Jin et al., 2017). Humphrey et al. (2019) demonstrated the spatially heterogeneous impact of urban vibrancy on safety. Moreover, Yu et al. (2022b) in a comprehensive analysis encompassing various facets such as accessibility, livability, diversity, and human activities, discerned that the progression of urban vitality can significantly contribute to reducing air pollution and PM_{2.5} concentrations. The observed phenomenon may be elucidated by the enhancement of accessibility and the diversification of land use. The improvement in accessibility can promote walking and cycling activities (Li et al., 2022d), thereby reducing the dependence on motorized vehicles, especially private cars, and consequently mitigating emissions of pollutants, such as nitrogen oxides (NO_x) and particulate matter (Stone Jr et al., 2007). Concurrently, integrating residential, commercial, and recreational spaces within mixed land-use configurations can diminish the imperative for car commuting (Cervero, 1996), consequently reducing air pollution.

4.3. Data source analysis

The analysis revealed that scholars used objective and subjective approaches to urban vitality assessment. For the objective approach, researchers used different aspects of human activity. For subjective research, researchers focus on how people rate the vitality of space or neighbourhood (i.e, human perception).

Regarding the objective perspective, initially, surveys and observations estimated pedestrian volume and human mobility (Sung and Lee, 2015). These traditional methods were labor-intensive but grounded in direct interaction with human subjects, offering rich contextual and experiential insights—albeit with limited spatial and temporal scope. However, their conclusions often rested on small samples and case-specific generalizations, which constrained their applicability across different urban contexts. With the advent of urban big data, scholars have

employed social media platforms (e.g., Weibo, Facebook, and Baidu check-ins), mobile phone location data, and Wi-Fi signals (Fig. 7). We observed that mobile phone data became the most prominent source for urban vitality studies from 2019-2024, followed by social media check-ins and Baidu/Tencent heat maps. Additionally, specific movements within urban spaces, such as bicycles (Li et al., 2022c; Zhang et al., 2023a) and taxi movements (Wang et al., 2021), are considered essential indicators of urban vitality. While biking and walking contribute to an environmentally friendly urban environment and promote a healthier lifestyle, taxi rides create opportunities for conversations that facilitate communication and may relate to street business activities. Moreover, some scholars proposed that human interaction time also indicates urban vitality, as used in a study by Heng and Zhang (2021). Over time, scholars have employed more diverse urban big data to evaluate urban vitality. For example, beyond mobile phone and social media check-in data, pedestrian counts from street view images, videos, and sensors have been used to demonstrate urban vitality (Kim et al., 2023; Hu et al., 2024).

On the subjective side, scholars have measured urban vitality through surveys to capture human perceptions of vitality (Aletta and Kang, 2018; Kang et al., 2020a; Paköz and Işık, 2022; Liu et al., 2023a; Zhang, 2023). For instance, researchers enlisted participants to evaluate human activities and assign vitality scores before employing artificial intelligence for urban vitality assessment (Qi et al., 2020; Li et al., 2022e; Ma, 2023). Other studies have revisited Jacobs' criteria, applying them through structured built environment assessments (Delclòs-Alió and Miralles-Guasch, 2018; Chen et al., 2023a; Vidal Domper et al., 2023).

We have noted a significant evolution in the data sources used to assess urban vitality over time. Figure 7 depicts the evolution of data sources utilized for evaluating urban vitality within the context of big data from 2013 to 2024. Between 2013 and 2016, traditional data such as human

mobility surveys, pedestrian manual counting, and GPS-based travel behaviour studies predominated (Marquet and Miralles-Guasch, 2015a; Sung and Lee, 2015). The 2010s marked the beginning of the urban big data era, enabled by advances in sensing technologies, cloud computing, and the widespread adoption of location-aware devices. At the same time, global policies have evolved with the adoption of the United Nations Sustainable Development Goals in 2015- particularly Goal 11, which emphasizes inclusive, safe, resilient, and sustainable cities and values data-driven decision-making (Kellison, 2022). These developments catalyzed a shift away from traditional approaches toward computational frameworks prioritizing quantifiable indicators. In this context, from 2017 to 2018, there is a discernible transformation towards incorporating technology-driven data sources, including WiFi, video surveillance, and location-based services (Kim, 2018). This trend continues and expands significantly between 2019 and 2022, marked by a substantial increase in mobile phone data usage, social media check-ins, and smart cards, underscoring a growing reliance on real-time, high-volume data (Tu et al., 2020; Li et al., 2022b). From 2023 to 2024, we observed that the diversity of data sources becomes even more pronounced with the inclusion of Baidu heatmaps, street view images, service reviews, and traffic data (Ma, 2023; Wu et al., 2023a). This evolution highlights the transition from conventional, manually collected data to sophisticated, automated, and diverse data, enhancing the accuracy and richness of urban vitality assessments. However, this transition also poses a risk of simplifying the concept of urban vitality to a function of density, diversity or movement alone, potentially overlooking the nuanced living experiences and differentiated access that characterizes vibrant urban life.

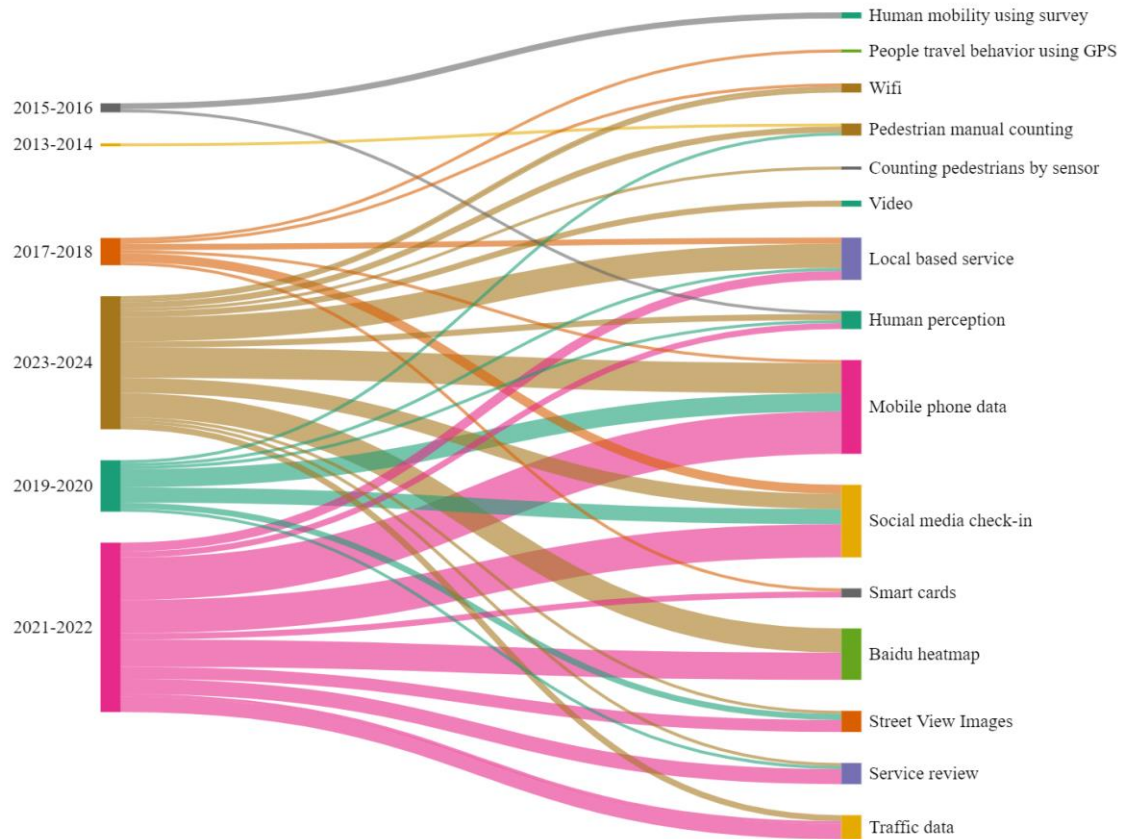


Figure 7. Variation of urban vitality data sources over the years.

4.4. Methods used in urban vitality study

Most urban vitality evaluation studies adopted a quantitative approach (i.e., 64 of 68 papers) based on geospatial big data. The final vitality maps were typically obtained by accumulating indexes of variables using spatial analysis, such as spatial overlay and spatial join (i.e., Paköz et al. (2022)). One-quarter of the papers utilized fishnets to estimate urban vitality at the micro-scale, with a 1 km grid being the most common choice, facilitating integration and visualization. The 1km grids and blocks are typically preferred when analyzing coarser data sources such as cell phone signals, check-in data, or aggregated census data (Yue et al., 2017a; Wang et al., 2022f). These larger units reduce data variability and smooth out small-scale anomalies, making them ideal for integrating multiple dataset such as land use and socioeconomic indicators. However, this

aggregation can potentially obscure disparities within neighborhoods and the influence of urban form at the finer scales. Grid analysis at a finer scale, such as 100m and 200m, often captures high-resolution spatial variability at the human scale. These analysis units are suitable for assessing intricate street-level characteristics such as streetscape design and pedestrian activities (e.g., through street view images) (Liu et al., 2020; Chen et al., 2023b; Doan et al., 2025). However, these spatial analysis units also face higher computational cost and biased outcomes due to sparse observations at small scales (Zhang et al., 2022c). Yet, few studies explicitly justify their choice of scale or conduct sensitivity analyses to assess how spatial resolution may affect the outcome. This calls for more critical examination of the modifiable areal unit problem (MAUP) in vitality research, where different grid sizes can lead to different conclusions (Xu et al., 2014).

Regarding weight, most documents assumed the same influence of variables on urban vitality. Only a few papers used the analytic hierarchy process (AHP) and entropy (Delclòs-Alió et al., 2019; Fuentes et al., 2020; Dong et al., 2021; Gong et al., 2023) to determine variables' weight in measuring urban vitality among 68 articles. A limited number of studies combined human perception of the urban environment and machine learning in urban vitality assessment (Qi et al., 2020; Li et al., 2022e; Ma, 2023). In contrast, only a few papers used a qualitative approach to explore vitality characteristics. The main qualitative approaches were a questionnaire survey and observations (Sung et al., 2013; Delclòs-Alió et al., 2019) to explore people's vitality evaluation. For example, Gómez-Varo et al. (2023) recruited seven participants and provided them with a prompt: "Identify the features and spaces that contribute to neighborhood vitality." Participants were asked to take photographs in response to this prompt and later took part in a group discussion where they presented their photos, explained the stories and meanings behind

them, and reflected on how these images relate to the broader concept of urban vitality (Coemans et al., 2017).

Various statistical models have been used to explore the relationship between urban vitality and other factors (Table 2). The most used models are the ordinary least squares (OLS) and spatial regression models. Among 123 articles, 58 used the OLS model to test the relationship between variables. However, the OLS model does not account for the geographical distribution of variables or spatial autocorrelation bias (LeSage and Fischer, 2008). Researchers, therefore, used other models, such as the Lagrange multiplier (LM) or geographically weighted regression (GWR). While SLM and SEM methods were used 15 and 13 times, respectively, GWR appeared 26 times (GWR: 13; geographical and temporal weighted regression - GTWR: 8; multiscale geographically weighted regression - MGWR: 5). In addition, some scholars are concerned about local interactions and the spatial pattern between urban vitality and other variables by using Moran's I. Specifically, Moran's I was used 15 times in studies to identify spatial patterns of vitality and spatial autocorrelation. Analysis of variance - ANOVA, Pearson's correlation, and chi-square were less frequently used to explore relationships.

Additionally, few studies acknowledge the complex relationships related to urban vitality (Xiao and Liu, 2023), so they have employed modern techniques such as gradient-boosting decision trees (Xiao et al., 2021; Han et al., 2023; Xiao and Liu, 2023), LightGBM (Chen et al., 2021), and random forest (Niu et al., 2022a) to gain further insights into the nonlinear relationships between urban vitality and its determinants.

Table 2. Methods used in urban vitality relationship studies

	OLS	SLM	SEM	GWR	GTWR	MGWR	Moran's I
Frequency	58	15	13	13	8	5	15

5. Discussion on limitations and research opportunities

Figure 8 presents a roadmap highlighting key urban vitality research gaps by linking them to prioritized research directions and appropriate methodological approaches. The framework underscores the importance of comparative studies across diverse geographical and urban contexts, the integration of temporal dynamics into vitality measurement, and the need to explore urban vitality relationships within broader sustainable development discourses. Research dimensions address pressing interdisciplinary gaps with broad societal impact and high potential to inform inclusive urban policies such as climate change, social inequity, urban resilience, and environmental issues, which are ranked the highest priority. These are fundamental problems that cities are already facing, having a large-scale influence on the urban system and are acknowledged as a key goal of the present urban agenda. (Johnston, 2016). The lower priority group is given to more context-specific. By aligning emerging technologies—such as IoT, digital twins, deep learning, and machine learning—with specific research gaps, Figure 8 aims to guide future studies toward more comprehensive, scalable, and policy-relevant contributions to vibrant urban futures. The following sub-sections provide a point-by-point discussion of current limitations and future research opportunities.

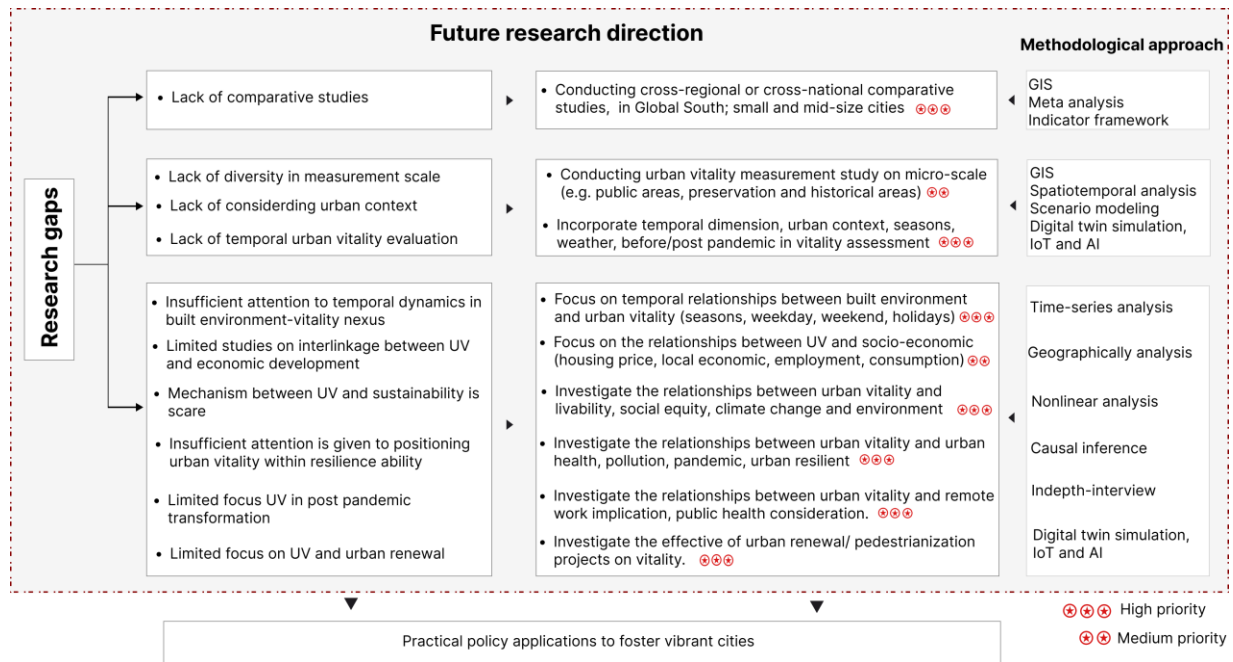


Figure 8. Future research directions and methodological approach

5.1. Research areas and comparative studies

It is noteworthy that existing studies predominantly concentrate on the urban vitality of megacities, with primary case studies conducted in cities in China, Spain, Turkey, and South Korea (Figure 1). These cities are characterized by high urbanization levels, elevated population density, intensive development, and extensive urban infrastructure. However, there is a conspicuous dearth of research exploring the vitality of developing cities. This imbalance may lead to an incomprehension of urban vitality across diverse contexts. Empirical evidence suggests that the dynamic processes and determinants of urban vitality in developing cities can significantly differ from those observed in developed cities in Europe and Asia due to distinct historical urban morphologies and territorial spatial planning policies (Yue et al., 2021; Wang et al., 2022e). Conducting on-site research in developing areas is crucial for addressing global inequalities and informing sustainable urban design. Hence, we encourage scholars to extend their research focus to encompass small and mid-size cities and exurban areas experiencing emerging public transport infrastructure and where walking may not be prioritized. Such investigations hold the potential to

provide additional insights into the understanding of urban vitality and how developing cities can generate impetuses for sustainable urban development.

Comparative studies exploring urban vitality dynamics across countries and continents have been relatively scarce (Zeng et al., 2018; Yue et al., 2021). More comparative studies on the urban vitality dynamics across continents are needed because the different cultural backgrounds and planning complexities inherent to different countries can significantly influence urban vitality patterns (Yue et al., 2021). For instance, the high housing and workspace density observed in the old urban cores of Chinese cities, shaped by traditional planning practices, may contribute to diverse conditions for urban vitality not found in newer urban areas. More comparative studies are necessary to comprehensively understand urban vitality by examining differences and similarities across continents. This global perspective facilitates the identification of common themes, distinctive qualities, mixed findings, and contextual elements influencing urban vitality. Such studies contribute to a nuanced comprehension of universal principles and region-specific variations.

5.2. Future research themes

5.2.1. Potential studies on urban vitality measurement

Urban vitality has been estimated at various scales, including streets, neighbourhoods, villages, districts, and cities, both during the day and at night. However, there is a noticeable lack of studies assessing vitality at a smaller scale. While macro-scale analysis provides an overarching view of urban vitality, it lacks the granularity to capture nuanced characteristics. In contrast, micro-scale analysis offers a meticulous examination of city spatiotemporal dynamics (Kim, 2020b), allowing for a detailed exploration of how vitality changes over different times or areas. Insights gained at the micro-scale are particularly significant for urban planners and policymakers, providing actionable information for targeted interventions and policies tailored to specific

challenges. Therefore, future studies could focus on micro-scale urban vitality (e.g., public areas, preservation or historical areas) to identify areas that need planning interventions.

Researchers have used different data sources to measure urban vitality, but more research is needed to compare the efficiency of these aspects. It remains unclear whether significant differences exist when using only mobile phone data versus human presence data. Future studies should compare urban vitality measurements using various datasets and test the reliability of these evaluations.

Current research focuses on estimating urban vitality but often overlooks the assessment biases caused by urban functions and the opportunities for human interactions. Urban functions, fundamental to urban vibrancy, are often considered in urban planning (Hall and Pfeiffer, 2013). Differences in urban function can lead to different built environments and social interaction characteristics. Heterogeneity in how different demographic groups may experience cities can cause gender segregation due to differences in inhabitants' preferences (Collins et al., 2023). Understanding a city's unique functions and demographic characteristics provides context for evaluating and interpreting urban vitality. Further research should employ comparative designs and cluster and typological analyses to reveal how diverse functional zones and population groups experience and shape urban vibrancy. These approaches enhance the robustness and contextual sensitivity of urban vitality assessments due to differences in zone function and demographic features across different urban settings.

Despite the evident changes in human activities across seasons, prior research has largely overlooked this aspect. Future studies should incorporate temporal dimensions to examine urban vitality across various seasons and temperature conditions for a more comprehensive understanding. Numerous studies have demonstrated the influence of climate and weather on

human activities, particularly walking (Aultman-Hall et al., 2009; de Montigny et al., 2011). Temperature significantly shapes outdoor activities, pedestrian movement, and overall urban behavior. For instance, high temperatures often lead people to seek shade or stay indoors with air conditioning, impacting street activities (Lin et al., 2012; Li et al., 2022a). Comparing urban vitality under varying temperature conditions can unveil patterns and preferences, contributing to developing more adaptable and resilient cities. Future research should utilize multi-temporal geospatial and IoT data to examine how vitality patterns shift across seasons and temperature gradients, given the well-documented influence of temporal factors on vitality and social interaction.

5.2.2. Potential studies on urban vitality relationship theme

5.2.2.1. Temporal dynamics in the built environment-vitality nexus

We discovered during our review that each theme category's size is not proportionate to innovation. For example, although there are many studies on the connection between the built environment and urban vitality, many of them are essentially duplicates or provide only slight improvements. Therefore, the relationship between these two dimensions needs further investigation. While urban vitality is dynamic, changing hourly and daily, the built environment remains relatively static over short periods. Exploring how the built environment influences vitality across different times of the day, days of the week, and seasons can provide valuable insights into the temporal dynamics of urban life. Advances in computational capabilities and artificial intelligence technologies offer promising solutions to explore these challenges. For example, deep learning can process complex street videos in real time, allowing researchers to observe human mobility and their interaction with the built environment. Empirical models can also contribute to the built environment-vitality nexus using high-resolution geotagged big data

collected across different times and days. Time-series models and spatiotemporal machine-learning approaches can reveal how built-form features interact with fluctuating activity patterns.

5.2.2.2. Interlinkages between urban vitality and economic development

While the association between urban vitality and the built environment has gained much attention, the impacts of urban vitality on economic factors are less explored. Economic aspects such as marketization, resident income growth, real estate, industry, and commerce development are directly tied to urban vitality (Barreca et al., 2020b; Lan et al., 2020). Some economic aspects needing further investigation include house prices (Lu et al., 2019; Li et al., 2022d; Liu and Shi, 2022), GDP (Jia et al., 2021), employment rate (Kang, 2020), retail consumption (Lan et al., 2020; Liu and Shi, 2022). For instance, Long and Huang (2019); Jia et al. (2021); Gómez-Varo et al. (2022) found a positive impact of average house prices on urban vitality. However, high house prices might also drive people to seek more affordable areas (Tsai, 2018; Yu and Cai, 2021), potentially reducing overall vitality. Understanding the complex relationship between urban vitality and economic aspects can help develop more inclusive and equitable cities, ensuring access to affordable housing, public services, and economic mobility. Further studies can examine the bidirectional and complex relationship between urban vitality and economic factors by employing econometric modeling approaches such as structural equation modeling and threshold effects models to detect non-linear and context-dependent effects. These approaches can assess how vitality influences—and is influenced by—indicators such as housing prices, employment, GDP, and retail activity.

5.2.2.3. Urban vitality and sustainability: liveability, social equity, demographic shift and climate change

Within the discourse of broader urban studies, very few studies have explored urban vitality in relation to livability and social inequity. While vitality highlights activity intensity, density, and diversity, liveability encompasses housing affordability, safety, environmental quality, and access to services (Gough, 2015). These two concepts may converge in some aspects, but they could also diverge. For example, vibrant commercial areas may create noise, congestion, and expensive expenditures that reduce residential satisfaction and liveability. In line with the nonlinear threshold effect mechanism, we posit that although urban vitality contributes positively to economic dynamism (Chen et al., 2022b), excessively high levels may yield negative externalities, including gentrification, strain on infrastructure, and environmental degradation (Yang et al., 2023a), directly influencing livability. Surges in population density and mobility can overburden transport systems and contribute to pollution, reducing liveability, as Bramley et al. (2009) have indicated that attachment feelings, residential satisfaction and stability, safety, and environmental quality are higher in lower-density areas. For example, in the Yangtze River Delta, intensified vitality has coincided with overcrowding and air quality deterioration (Fang et al., 2021b; Liu et al., 2022a), which can reduce livability. Therefore, urban vitality should be critically assessed, rather than simply maximized, to ensure its benefits are equitably distributed and do not undermine livability.

Regarding social equity, vitality initially may attract capital and jobs (Liu et al., 2022a), but rising housing prices in highly vital areas may exclude lower-income groups (Plantinga et al., 2013; Zhou and Chi-Man Hui, 2022), thus exacerbating inequality (Kirdar and Cagdas, 2022). Projects aimed at promoting vitality, if not inclusively designed and effectively planned, can potentially exacerbate social disparities. This is because individuals from disadvantaged groups might be displaced or denied access to the benefits generated by urban vitality enhancement initiatives (Ward et al., 2019). As a result, neighborhoods with intense vitality may favor affluent

social groupings while excluding those from lower socioeconomic backgrounds (Delclòs-Alió and Miralles-Guasch, 2018; Connolly, 2019). This suggests that designing for vitality may paradoxically promote exclusionary practices like neighborhood gentrification. Though it is crucial for advancing social justice, the connection between urban socioeconomic disparity and the conditions necessary for urban vibrancy is not well studied.

To address these research gaps, an integrated methodological framework integrating quantitative spatial analytics and qualitative participatory approaches is essential. Structural equation modeling and machine learning can be employed to quantify nonlinear relationships (e.g., threshold effects) between vitality metrics (density, diversity) and livability indicators (housing costs, pollution), while controlling for socio-economic inequities (Nedungadi et al., 2024). And, participatory approaches, including public engagement and cognitive mapping with marginalized communities, will critically examine the lived experiences of exclusion in high-vitality zones (Tiwari et al., 2025). This enables both macro-level pattern identification and micro-level experiential insights, advancing theoretical understanding while informing vitality sustainability frameworks.

Many developed and developing countries are experiencing rapid demographic changes, like aging and depopulation, that have profound implications for urban vitality, yet this dimension remains relatively underexplored in current urban studies. As the demographic composition of cities changes, so do social interaction and mobility patterns, shaping urban life's vibrancy (Urbanek et al., 2023). For instance, people aged 60+ have been proven to have lower mobility and interaction than younger people, indicated by fewer trips and travel distances (Currie and Delbosc, 2010). Therefore, an area with an aging population may lead to reduced nighttime activity and increased demand for accessible public spaces and services, while youth outmigration can

result in decreased cultural and economic dynamism (Meyer and Leibert, 2021). Future studies should investigate how specific demographic changes influence the spatial-temporal distribution of vitality, particularly in different urban contexts. Researchers can uncover trends and complicated relationships by integrating demographic data with mobility patterns, social media activity, and land use via spatial-temporal analysis. Scholars can also integrate qualitative methods such as interviews and focus groups to capture the lived experiences of diverse groups. Such studies can inform urban policies that accommodate demographic diversity and foster equitable, vibrant cities that respond to evolving population dynamics.

Despite the recognized importance of urban vitality for sustainable development, its relationship with climate change issues remains underexplored. Climate change intensifies environmental pressures in urban areas through rising average temperatures, more frequent and severe extreme weather events (e.g., heatwaves, storms, heavy rainfall), sea level rise, and prolonged droughts (Masson-Delmotte et al., 2021). These phenomena directly affect the quality and usability of public spaces, which are core to maintaining urban vitality (Elmqvist et al., 2019). For example, extreme heat driven by climate change reduces thermal comfort in outdoor environments, making streets, plazas, and parks less attractive or unsafe for public use. This results in decreased pedestrian activity and weaker social interaction, which are essential for a vibrant neighborhood. Additionally, Cutter et al. (2018) emphasizes that a significant consequence of climate change is flooding, which often affects urban areas that were once considered secure. This phenomenon threatens the vitality of cities because shelters become more vulnerable and entire regions may become uninhabitable. A comprehensive understanding of the effects of disasters and climate change on urban vitality is crucial for introducing appropriate short, medium, and long-term solutions for populations that can be potentially affected (Yamamoto et al., 2017). To fill this

gap, future studies can take a longitudinal study, systematically tracking urban vitality alongside climate change events (temperature extremes, floods) to analyze adaptation strategies and tipping points where extreme events degrade vitality. Besides, scholars can integrate semi-structured interviews to unpack lived experiences, revealing people's adaptive strategies to climate change.

5.2.2.4. Resilience thinking in the context of urban vitality

Although cities are vulnerable to various environmental, economic, and social shocks and stresses, such as economic downturns and pandemics, a notable gap exists in urban vitality studies incorporating resilience and adaptability. The resilience concept introduces a temporal and systemic perspective, highlighting how cities absorb, adapt to, and recover from disturbances such as pandemics, climate events, or economic shocks. In this context, vitality reflects a city's ability to sustain and regenerate everyday urban life under stress (Amirzadeh et al., 2022). While previous studies have explored resilience in terms of infrastructure, public health, or governance (Hu et al., 2022), the resilience of urban vitality—encompassing social interaction and public space use—remains underexplored. For example, during the pandemic, the widespread cancellation or online transition of large events resulted in a general decline of urban vitality, and areas that had higher pre-pandemic vitality levels experienced more significant drops (Liu et al., 2023d). Many cities continue to face challenges and have not fully recovered from the effects of the crisis. The economies of many cities were variably affected and took years to recover to their original states (Hayakawa and Keola, 2021). In a post-pandemic period, a certain percentage of the population relocated from cities (Paköz and Işık, 2022), or there was an increasing migration trend from city centers to suburbs (Gallent and Madeddu, 2021), reducing urban vitality in city centers. The questions such as “How does urban vitality alter during the pandemic? Has vitality returned to what it was before the outbreak? What factors contribute to urban vitality resilience?” still need to

be further investigated. Understanding the impact of pandemics or economic downturns on urban vitality is essential for developing resilience strategies. Cities can enhance their adaptive capacities by identifying vulnerabilities, preparing for future crises, and informing adaptive urban design strategies (Allam and Jones, 2020). City authorities may need to reconsider design aspects such as public spaces, transportation, and density to bolster adaptability and ensure that urban areas remain vibrant and livable even during challenging times (Banai, 2020). Therefore, future studies can focus on resilience and adaptability strategies in urban systems to maintain vitality (Amirzadeh et al., 2023). Methodologically, this could involve longitudinal analyses of urban mobility and interaction patterns using diverse data sources before, during, and after disruptions. Comparative case studies across cities with differing resilience performances can also help identify which spatial, policy, and design attributes support faster and more inclusive vitality recovery. Integrating spatial econometric models with resilience indicators, employing machine learning and difference-in-differences approaches to detect tipping points and causal inference in vitality fluctuations, may offer new insights into the adaptive dynamics of urban vibrancy.

5.2.2.5. Urban vitality in post-pandemic transformations

The COVID-19 pandemic has reshaped perceptions of urban vitality, creating a tension between vibrant public spaces and emerging public health considerations. During the COVID-19 pandemic, the infection rate increased dramatically, particularly in densely populated areas where elevated levels of social interaction in public spaces heightened the risk of exposure to the virus. Numerous preventative measures significantly limited people's freedom of movement, travel, and social interaction, thereby altering typical social and economic activity patterns and affecting urban vitality and public health (Dong and Bouey, 2020). In the post-pandemic period, a segment of the population remains psychologically sensitive to crowded environments with high levels of social

interaction, often as a lingering effect of COVID-related trauma (Kainthola et al., 2021; Yang et al., 2022a). Yet critical knowledge gaps persist, including defining “safe” density thresholds and ensuring designs serve psychological safety. Additionally, following the gradual alleviation of the epidemic, the government's post-epidemic program included a plan for economic and social rehabilitation (Nian et al., 2020). Researchers have looked at various factors to fully comprehend these shifts, such as urban government, traffic, and population density. There is, however, little study on urban vitality after the pandemic, raising important questions such as how pandemic-induced behavioral changes, such as aversion to crowded spaces, persist over time, and how urban design strategies can foster resilience without compromising vibrancy? Could the pandemic be a catalyst for decentralization and vibrant cities?

Furthermore, much activity during COVID-19 moved from offline to online, which continued to some extent even after the epidemic was mostly contained. With most people working and studying from home, commuting and private vehicle use plummeted, and cities experienced rapid declines in traffic (Sharifi and Khavarian-Garmsir, 2020). These virtual pursuits also reflect a certain level of urban vibrancy. It may be possible to assess urban vitality more objectively by considering online economic activity (Li et al., 2024b). Understanding how a pandemic affects people's behaviors and social activities can be achieved by researching how urban vitality varies during the outbreak. In order to better respond to future epidemics, this can aid in developing more efficient public health and emergency management plans. By comprehending these shifts, governments can also create appropriate social and economic policies to support urban development and recovery (Wu et al., 2023b).

Future studies should adopt longitudinal and mixed-methods approaches to capture both the temporal dynamics and qualitative shifts in urban vitality during and after pandemics.

Integrating spatial analysis with real-time data sources—such as mobility data, social media activity, and service usage records—can help map differential impacts across urban zones. Comparative case studies between high-density and low-density environments and offline versus online activity patterns are essential to uncover how various urban forms respond to crisis-induced disruptions. Additionally, scenario-based simulations using machine learning and agent-based modeling can explore the implications of different urban design and policy interventions on long-term vitality and resilience.

5.2.2.6. Urban renewal and urban vitality

Urban renewal initiatives have become essential tools for refining urban layouts and spatial structures while boosting vitality. These projects typically involve the functional transformation of areas suffering from poor environmental quality, inefficient land use, or outdated functions misaligned with development needs (Zheng et al., 2014). Urban renewal is also recognized as a viable solution for addressing urban decay, enhancing the living environment, and achieving various socio-economic goals (Zheng et al., 2014). Urban renewal projects can alleviate community health risks, promote efficient land use by upgrading infrastructure, repurposing spaces, and improving public amenities (Ho et al., 2012), and generate vibrant, inclusive environments that attract businesses, residents, and cultural activities. Therefore, urban renewal can stimulate the city's intrinsic vitality and foster an endogenous development dynamic.

While the renewal of public spaces plays a central role in enhancing spatial vitality, the actual efficiency of such interventions in fostering urban vibrancy remains underexplored. Existing studies often face limitations due to the lack of high-resolution spatial and socio-behavioral data, which hampers the ability to systematically evaluate specific changes in spatial elements (Fan et al., 2023). Therefore, future research should prioritize developing interdisciplinary evaluation

models that assess the effectiveness of spatial enhancements (e.g., pedestrianization project, public space) to social interaction and vibrancy. To address this gap, future studies may consider employing quasi-experimental designs such as synthetic control or difference-in-differences methods and longitudinal big data sources to establish causal relationships, providing insight into the effect of urban renewal on vitality.

5.3. Limitation of current data sources

Measuring urban vitality has historically been a complex task (Azmi and Karim, 2012) that has often overlooked the intensity of human activity (Montgomery, 1998). Some studies have collected human activity data through conventional methods such as observation, surveys, and questionnaires to address the limitations of static data. While such methods provide detailed demographic information such as age, gender, and occupation, their implementation cost is relatively high and unsuitable for large-scale studies (Wang et al., 2022d). The prevalence of open urban big data has revealed human mobility in public spaces (Long and Liu, 2016; Wang et al., 2022d; Chen et al., 2023b) through data sources such as mobile phones, social networks, smart cards, banking, taxis, and buses. With the ability to determine human position, GPS data is considered an adequate proxy to represent human interaction in spaces (Jacobs-Crisioni et al., 2014). However, only using open big data has some limitations (see Table 3).

Considerable limitations of urban big data have appeared in previous studies. Patrizia Sulis et al. (2018) used Twitter and transportation cards as proxies of human activities, but such data cannot fully represent the activities of elderly, young, or poor people (Giles, 2012). In addition, smart card data only records the start and end of railway and metro journeys and fails to capture interactions among people in street spaces. Moreover, open data sources (e.g., Baidu, Tencent, Weibo, and mobile phone data) that are widely used in quantitative studies in China have limitations in proxying human mobility and cannot differentiate indoor versus outdoor activities

or surrounding environments. Another common challenge with using human big data is data privacy, as analyzing human mobility data can extract behaviors, habits, and lifestyle information, thereby compromising human privacy (Renso et al., 2013). As a result, mobile phone companies tend to be hesitant to share data that contains personal information. These barriers make it difficult for researchers to obtain the necessary data to study urban activity patterns despite the potential benefits of using this data type. Therefore, future studies should take protective measures, such as who-where-when model (Smolak et al., 2020) and using semantics instead of raw data (Lee et al., 2011; Barak et al., 2016) when working with human-related data.

Furthermore, the assessment of urban vitality increasingly utilizes street-view imagery (SVI) data to extract pedestrian volumes, analyze built environment characteristics, and evaluate street-level spatial qualities. While the accessibility and scalability of SVI offer clear advantages for automated and wide-scale urban analysis (Kang et al., 2020b), its application also raises critical concerns regarding data quality, spatial representativeness, and methodological bias, which can significantly undermine the validity of vitality assessments. One major limitation is temporal inconsistency (He and Li, 2021). SVIs are captured at different times, leading to a mismatch between the imagery on the same street (Ma, 2023). Pedestrian counts in SVI are inherently dynamic and vary significantly depending on when the SVI image was taken. This issue is especially pronounced when imagery is outdated or collected imagery from various seasons is used in the same analysis. For instance, vitality indicators such as pedestrian presence may be lower in winter imagery (He et al., 2020; Doan et al., 2025). SVI metadata frequently omits specific temporal markers such as the day of the week or the time of day. These dimensions are particularly salient in urban vitality research, as street-level vibrancy often fluctuates by hour (e.g., peak hours vs. non-peak hours) and by day (e.g., weekdays vs. weekends) (Wang et al., 2022f). The absence

of such temporal detail may result in systematic misrepresentation of activity levels, thereby undermining the reliability of vitality assessments (Wang et al., 2022c). Geographic coverage also remains uneven. While services like Google street view imagery provide dense coverage in urban cores, they are often limited to major roads in smaller towns and are frequently absent in rural and lower-income regions (Mahabir et al., 2020). This skews the spatial focus of urban vitality research toward well-developed urban centers, excluding marginalized or underrepresented areas and limiting the generalizability of findings (Wang et al., 2022c). In addition, inconsistencies in data acquisition methodologies further complicate analytical robustness. Imagery is typically captured from commercial and volunteer-operated vehicles with varying camera specifications, mounting angles, and positional heights. Such heterogeneity affects image composition and viewpoint, resulting in inconsistent representation of key street-level features. Access constraints and viewpoint limitations hinder the full capture of pedestrian environments. Many SVIs are collected by vehicles, making narrow alleyways, pedestrian-only streets, and informal settlements inaccessible. For example, Jiang et al. (2022b) noted that SVI captured from cars may fail to represent human-scale perspectives or cover diverse spatial settings, which compromises the accuracy of pedestrian-related and human perception analyses. Image quality and clarity are another concern. Environmental factors such as poor lighting, obstructed views, and weather conditions often degrade image usability (Li et al., 2018; Biljecki and Ito, 2021). Finally, the sky, roads, trees, buildings, and pavement deform in panoramic images; the calculations of the sky, road, tree, building, and pavement ratios are not realistic in segmentation analysis. These issues seriously challenge computer vision models in urban vitality studies.

To address this limitation, urban vitality studies can combine street view images from multiple sources (e.g., location services and social networks) to enhance street data (Zhang et al.,

2020b). Images captured by devices mounted on bicycles, motorcycles, and pedestrians can also help capture many scenes not captured by cars. Table 3 presents the advantages and disadvantages of data in measuring urban vitality.

Table 3. Advantages and disadvantages of different data types

Author	Data	Advantage	Disadvantage
(Sung and Lee, 2015; Seong et al., 2023)	Pedestrian volume by observation	Number of people can be recorded with high accuracy	- High cost of data collection - Can not apply in macro-scale
(Yue et al., 2017a; Li et al., 2022b)	Activities of mobile phone users in an hour, day	Provide spatial-temporal granularity, spatial temporal rhythm	Not revealed to the public, difficult to obtain this data
(Wu and Niu, 2019; Kim, 2020a; Kim, 2020b)	Mobile phone users' location	- Provide spatial temporal granularity, spatial-temporal rhythm. -Record huge amount of position data.	- Not easy to access. - Not provide the people activities. - Low accuracy
(Wu et al., 2018b; Jiang et al., 2022b)	Handheld GPS	- Good positioning and anti-interference ability	High price
(Chen et al., 2019; Lang et al., 2022; Niu et al., 2022a; Ouyang et al., 2022; Wang et al., 2024a)	Social networks such as check-in, comment on Facebook, Twitter, Tencent, Weibo, e-commercial platforms	Can reflect where attracts more people and more contribute to urban vitality	Ignore people who do not use social media platforms, such as younger and elderly people
(Kim, 2018)	Wifi connection	-Accuracy estimates the number of people accessing to the system -Provide another aspect of visual vitality	- Can not reflect people use their own data, such as 4G, 5G, or private wifi. - May not cover young and elderly people
(Zeng et al., 2020; Dong et al., 2021; Zhang et al., 2022c)	Taxi drop-up point/ Bike sharing/ smart card	Record the trip in detail	Lack of pedestrian activities Only reflect terms of destinations (start and end points)
(Xia et al., 2020; Zhang et al., 2020a; Zhang et	Nighttime light data	-Free to access - Reveal urban expansion pattern	- Cannot reflect people's activities, interaction - Low spatial resolution

Author	Data	Advantage	Disadvantage
al., 2022b; Zhang et al., 2024)		- Can be used for nighttime vitality	
(Ye et al., 2018; Xia et al., 2020; Xiao and Liu, 2023)	Small catering/ food facilities	Reflect the crowd because it is in easily accessible.	Small catering only reflects a specific aspect of vitality
(Qi et al., 2020; Li et al., 2022d)	Street view images	- Free to access - Reveal the surrounding environment and human	- Street images are captured at different times by different people - Can not reveal pedestrians during peak/non-peak hours

Emerging technologies such as the Internet of Things (IoT), autonomous systems, and digital augmentation are revolutionizing the measurement of urban vitality. The Internet of Things (IoT) enables the widespread deployment of sensors across urban spaces, providing real-time data on environmental variables (e.g., air quality, temperature, noise), infrastructure performance, and human activity patterns (Wang et al., 2022f). For instance, initiatives such as Chicago's Array of Things (AoT) and Seoul's Data of Things (S-DoT) exemplify how sensor data can provide insights into public space utilization, informing urban planning and policy decisions (Kim, 2025). These advanced techniques now allow stationary cameras connected to IoT networks to sequentially process image frames, deriving vehicular speed, mobility patterns, and even human interaction dynamics (Yang and Pun-Cheng, 2018; Dai, 2019). Autonomous systems, including self-driving vehicles and drones, contribute to urban vitality by enhancing mobility and accessibility (Medina-Tapia and Robusté, 2018). These technologies can optimize traffic flow, reduce congestion, and improve safety, thereby encouraging more active use of urban spaces (Zheng et al., 2020). Moreover, autonomous systems, such as self-driving vehicles and drones, further influence urban vitality by reshaping mobility and logistics (Bakirci, 2025). Meanwhile, autonomous delivery and transport services help strengthen connections between urban dwellers and local businesses. These technologies promise to optimize traffic flow, reduce congestion, and enhance accessibility and

safety factors that are key to dynamic urban life. Digital augmentation technologies—including augmented reality (AR) and digital twins—offer new ways of interacting with urban space. Digital twins allow planners and researchers to simulate urban systems under different scenarios, improving decision-making and resource management through their ability to provide engaging, real-time visualization and simulations of the urban environment (Caprari et al., 2022). For instance, by simulating urban scenarios, city planners can anticipate and mitigate potential issues, ranging from traffic congestion to environmental degradation (Mylonas et al., 2021). Simultaneously, AR tools enrich public spaces with real-time information, fostering engagement, education, and participatory experiences that support social vitality (Imottesjo and Kain, 2018).

While these technologies present significant potential, their integration must be cautiously approached. Overemphasis on technological solutions can overshadow essential human and cultural dimensions of urban life. Therefore, the focus should not solely be on technological advancements but also on ensuring that these innovations enhance sustainability and livability (Allam and Dhunny, 2019). A balanced approach incorporating cultural, metabolic, and governance considerations is crucial for successfully applying emerging technologies to foster urban vitality.

Future research should use multidimensional urban data sources to fully comprehend the distribution patterns of urban vitality. This is because data from a single source is insufficient to fully portray the multidimensionality of an urban place due to the diversity of social, economic, and cultural activities that occur there (Huang et al., 2019). For example, while cell phone signal density may provide insights into the agglomeration of people, its capacity to discern the overall abundance of urban activities remains limited (Akinci et al., 2022). Multidimensional urban data can describe the multidimensionality of urban vitality, enabling scholars to comprehensively

understand the distribution pattern of urban vitality and its influence factors (Huang et al., 2019; Kang et al., 2020a).

5.4. Method limitations and potential methods

Urban vitality is influenced by various factors, some of which play a crucial role, while others are supporting factors for a comprehensive assessment. However, previous studies have often ignored the weight of these factors, and few studies have determined the factor's weight to measure urban vitality (Delclòs-Alió et al., 2019; Dong et al., 2021; Xia et al., 2022b). Some studies did not clearly describe how the variables' weight is calculated. Neglecting the consideration of weights can introduce bias and compromise the accuracy of the obtained results. A pressing need, therefore, exists to develop a set of weights for measuring urban vitality. Weight calculations can be determined by expert methods such as entropy, AHP, and analytic network process (ANP).

In urban vitality studies, deep learning algorithms (CNNs, ConvNets, and YOLO) are often used for image analysis (Yu et al., 2018). However, we found that some previous studies lacked technical details hindered reproducibility due to the absence of sharing of development materials, such as codes and trained models. These sophisticated models were used to interpret urban vitality by assigning scores akin to human evaluations (Kruse et al., 2021; Ma, 2023) and pedestrian detection (Li et al., 2022d). However, the assumptions and limitations of machine learning and deep learning warrant scrutiny. These models typically learn patterns from other large datasets, which may encode cultural and contextual biases (Wang and Biljecki, 2022). For example, an image labeled as "vibrancy" in one urban setting may not generalize to others with different architectural styles and social behaviors. Moreover, while CNNs excel at recognizing visible cues such as greenery, building facades, or pedestrian presence (Ma, 2023), they struggle to account for

less tangible but equally vital elements, like inclusivity, social cohesion, or perceived safety. One key issue is the "black-box" nature of deep learning models, which often makes it difficult to interpret how predictions are generated and the mechanism behind which features most influence urban vitality (Ramírez et al., 2021). This lack of transparency can undermine confidence in the models' outputs and introduce uncertainties into the results. In many cases, pedestrian detection results still require manual validation—human raters are often used to confirm the accuracy of predictions, as He et al. (2020); and Song et al. (2021) suggested that these tools are not yet fully autonomous or self-explanatory. Participation involvement, such as deep interviews and photovoice, can be integrated to address this limitation. In comparison to structured surveys and AI cognitive assessments using keywords such as “safe” or “vibrant,” Photovoice offers a deeper and more nuanced exploration of urban vitality. While citizen perception surveys or AI-based image analysis capture direct impressions, Photovoice reveals the underlying narratives, emotional attachments, and socio-spatial dynamics that shape people’s experience of vitality. It not only records what participants observed but also why they interpret it in such a way, bridging visual stimuli with lived realities (Erfani, 2021).

Methods used to explore the association between urban vitality and its determinants are diverse, such as OLS, GWR, MGWR, and local indicators of spatial autocorrelation (LISA). While the OLS model does not consider the spatial impact on error coefficients, GWR can enhance performance by estimating coefficients at distinct locations (Fotheringham et al., 2001; Zhang et al., 2020a). Technically, scholars suggest that MGWR has a more flexible scale than GWR, making it better at solving collinearity issues and producing better results (Fotheringham et al., 2017; Murakami et al., 2019). Specifically, multiscale geographically weighted regression uses a variety of error structures, including spatially correlated errors, which can improve the model’s

accuracy. Additionally, MGWR provides better predictions than GWR because it combines local and global parameters to generate predictions. However, the estimation of parameters in these methods relies on linear assumptions. Studies have also used LISA to test the spatial autocorrelation in urban vitality distribution (Barreca et al., 2020b; Tu et al., 2022) and LM to test lag and error in the spatial model (Botta and Gutiérrez-Roig, 2021; Yu et al., 2022b). Both LISA and LM are suitable for handling large datasets. By analyzing the spatial relationships between variables, they can reveal relationships that may be difficult to detect using other methods. While LISA is particularly useful for identifying hotspots and cold spots to understand the spatial distribution of factors in the urban environment, LM can determine whether the observed relationships are random or statistically significant.

It is important to note that most studies examining the relationships between urban vitality and its determinants are predominantly based on linear assumptions (Jiang et al., 2022b; Wu et al., 2022d). These methods operate under the assumptions of linearity, additivity, and homoscedasticity. They often oversimplify the multifaceted nature of urban dynamics. Traditional linear regressions also assume that the marginal effect of an independent variable remains constant across its entire value range (Grömping, 2009). As a result, they may either overestimate or underestimate the influence of independent variables, especially when the underlying relationships are nonlinear or involve interaction effects (Doan et al., 2024). This overlooks the possibility of threshold effects common in urban systems where, for example, increasing density may enhance vitality up to a point but become detrimental beyond it (Xiao et al., 2021; Han et al., 2023). Furthermore, these models struggle to capture synergistic or mediating relationships among variables, leading to mixed findings and inconsistencies in vitality literature (Xiao et al., 2021; Xiao and Liu, 2023).

In contrast, machine learning approaches offer significant advantages in this regard. These models can uncover complex, nonlinear patterns without requiring prior specification of functional forms (Wu et al., 2023c; Ming et al., 2024). They also allow for the identification of variable importance, interaction effects, and spatial heterogeneity, offering a more nuanced and potentially accurate understanding of how urban form impacts vitality (Xiao et al., 2021; Yang et al., 2021a; Tang and Ta, 2022). However, the application of these methods remains limited as their black-box nature can pose interpretability challenges, particularly in contexts where policy implications demand clear explanations (Xiao et al., 2021). Also, machine learning models often optimize for accuracy; without careful feature selection and model diagnostics, there is a risk of overfitting or misattributing correlations to causal drivers of vitality

Accordingly, future studies can (1) use new methods such as machine learning techniques to explore more non-linear relationships and to understand the influence level on urban vitality; (2) compare different techniques within a specific topic to observe the differences in their relationship; and (3) develop a new framework beyond linear modeling methodologies that covers the complex interplay between social, economic, and environmental factors that contribute to urban vitality. Additionally, it is important to consider how to apply the findings in practice and policy. This work requires a multi-disciplinary approach that draws on diverse perspectives, skills, and expertise to address complex urban challenges.

Finally, considering the temporal aspect, it is important to note that longitudinal studies are rare. Existing studies predominantly focus on establishing associations between urban vitality and various urban features, yet causal relationships remain largely unexplored. Uncovering causal relationships is a viable research direction. Causality studies are instrumental in comprehending how urban policies, interventions, and other events affect urban vitality levels. For example,

controlling differences between treated and controlled groups of samples before policy implementation facilitates the identification of the net effect of control policies on vitality (Tan et al., 2022; Wu et al., 2022c). This strategic approach empowers policymakers to discern the true impact of interventions. Future studies should delve into designing natural, longitudinal experiments to unravel the relationship between urban vitality and its determinants.

6. Conclusion

We have extensively reviewed geographical distribution, research themes, data and methods in urban vitality studies by examining 191 publications. Additionally, we mapped out the most prominent pattern in urban vitality studies before identifying the knowledge gaps and research opportunities for future work. Our analysis revealed a predominant pattern: a surge in publications over the last four years, with a concentration in East Asia and Southern Europe, particularly China. The primary focus of urban vitality studies includes measurement techniques and exploring the bidirectional relationships between urban vitality and other aspects of urban life. While the relationship between urban vitality and the built environment was well studied, its relationship with sustainability (social and environmental issues), urban renewal and urban resilience are far from enough. Notably, we found a significant transformation from conventional, manually intensive methods to sophisticated, automated, and diverse data collection techniques in the big data era. In terms of methods, overlay is prominent in vitality measurement, whereas OLS and spatial regression are commonly used in relationship studies. Studies on the nonlinearity mechanism and causality between urban vitality and its determinants are limited.

There are three main takeaways we highlight as the main research opportunities. Firstly, expanding the study regions to include developing cities in Asia, the Americas, and Africa will offer valuable insights into urban vitality across diverse contexts. Comparative studies on urban

vitality dynamics across continents will be particularly beneficial. Secondly, a thorough assessment of urban vitality should involve exploring its variations across seasons and scales, incorporating human-centric variables and behavioural aspects. We propose that further investigations into the relationship between urban vitality and environmental and social issues, such as livability, congestion, population shift, pollution, public health, and resilience, are essential. Thirdly, considering the nonlinearity mechanism, there is a need for in-depth exploration of the threshold effects of determinants on urban vitality, the impact of urban vitality on the urban environment to manage potential adverse effects, and promote resilient and adaptable urban systems to identify optimal thresholds for its maintenance. Further studies should explore the causality mechanisms by designing natural, longitudinal experiments and difference-in-difference models to shed light on how urban policies, interventions, and other events impact urban vitality levels. It is crucial to bridge the gap between research findings and practical policy applications to foster vibrant and resilient cities.

This review paper has limitations related to the type of input documents. The literature reviewed is limited to peer-reviewed publications indexed in WoS and published in English, which may result in geographical distribution bias and exclude articles published in other languages. In addition, this review does not include books and preprint papers, which may convey new directions not captured by this paper. Despite these limitations, this study provides a comprehensive overview of existing insights on urban vitality and its relationships, highlighting research gaps and future research directions.

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